



Master's Degree in Finance

Course of Empirical Finance

Roughly Speaking,
on the Empirical Texture of Volatility

Prof. Paolo Santucci de Magistris

SUPERVISOR

Prof. Giacomo Morelli

CO-SUPERVISOR

Simone Pisano

CANDIDATE

Academic Year 2024/2025

Contents

1	Introduction	3
2	Stylized Facts of Volatility	6
2.1	Volatility Clustering	6
2.2	Leverage Effect and Asymmetry	7
2.3	Persistence and Memory	8
2.4	Vol-of-vol and Microstructure Effects	10
2.5	Roughness	11
2.6	Scaling Laws and Universality	12
2.7	On Memory and Mean Reversion	13
3	Modeling Volatility	16
3.1	GARCH model	17
3.2	HAR-RV model	18
3.3	RFSV model	19
4	Empirical Analysis	24
4.1	Data	25
4.2	Parameter Estimates	31
5	Results and Conclusions	37

1 Introduction

Volatility stays at the center of modern finance, and it affects deeply many aspects of how we manage risk or price derivatives. Whether we are talking about allocating capital in a risk aware manner, or designing strategies to withstand turbulent periods in the markets, understanding how asset prices fluctuate is essential. In a sense, volatility acts as the “temperature” of financial markets: it does not move assets by itself, but it sets the stage for how sensitive those moves can be. More concretely, volatility tells us how uncertain we are about where prices will go, and this uncertainty shapes nearly every financial decision, from how to hedge a portfolio to how to quote the price of a vanilla option.

A solid grasp of volatility dynamics allows practitioners to estimate future risk in portfolios, adjust exposures, and design more effective trading strategies that respond sensibly to market regimes. On top of that, volatility enters directly into the pricing equations for a wide class of financial instruments. Getting its behavior wrong means mispricing risk, which is always a bad idea.

From a theoretical perspective, modeling volatility is not only about forecasting, it is about explaining. It helps reconcile stylized facts in the data that are otherwise hard to justify: the heavy tails in asset returns, the volatility clustering we observe during crises and calm periods alike, and the asymmetric way markets tend to respond to bad news versus good. These features are deeply ingrained in financial time series and cannot be ignored. In the past, and still today to a large extent, a lot of this modeling relied on autoregressive structures like the GARCH family, where the idea is simple but powerful: the variance we observe today is some function of what we have seen in last periods. These models, first introduced by [Engle and Bollerslev \(1986\)](#), became extremely popular because they work reasonably well and consistently across different settings.

As the availability of high-frequency data and more granular market observations increased, new techniques could be developed on finer time grids. GARCH models, while still widely used and useful, often struggle to reproduce the fine temporal structure of volatility observed in empirical data. In particular, they tend to understate the roughness of volatility paths, especially when we zoom in at very short time scales. These additions led to the development of the rough volatility, not as a purely theoretical construct, but as a response to concrete limitations of classical models. Bid-ask bounce, the discreteness of price changes and intraday seasonality were new problems to deal with as they contribute to what is generally referred to as microstructure noise. By accounting for these distortions, highly erratic intraday fluctuations, and long memory effects that extend beyond the reach of short-memory autoregressive specifications, rough models open the door to a more faithful representation of how volatility actually behaves in practice, not in an average sense, but path by path.

This thesis starts from this point: the idea that volatility, while central and heavily

studied, might not yet be modeled in a way that reflects its true nature, especially when we look closely. And if volatility is indeed rougher than previously thought, then it follows almost naturally that many of the tools we use might be missing something fundamental. This is the driving reason of this research, by encapsulating rough features in the volatility dynamics it can be possible to

This is the driving reason of this research, by encapsulating rough features in volatility dynamics, it becomes reasonable to confront classical models with new ones and to ask an important question: which description captures better the data we actually see? The aim of this thesis is therefore to conduct a systematic comparison between traditional frameworks, such as the GARCH family and realized volatility measures, and more recent approaches where log-volatility evolves as a fractional Brownian motion. The analysis is carried out under the historical probability measure, for reasons we will discuss later on.

The traditional approaches under analysis are Generalised AutoRegressive Conditional Heteroskedasticity (GARCH) model and Heterogeneous AutoRegressive Realized Volatility model (HAR-RV). GARCH(1,1) is the most famous model in volatility modeling because of many reasons, the first being its effectiveness even considering it is parsimonious in parameters and assumptions. This model does not lack evidence in its validity, grounded in years of academic research and practitioners' use. HAR-RV emerges from the econometric tradition of realized measures and aims at capturing the multi-level persistence of volatility by decomposing its dynamics in daily, weekly and monthly components, providing a flexible structure that resembles the behavior of market participants across different timeframes. Finally the Rough Fractional Stochastic Volatility (RFSV) model departs from both these frameworks by treating log-volatility as a process with a fractional Brownian Motion engine, with Hurst parameter $H < \frac{1}{2}$. This parameter encodes the degree of roughness of the simulated path, its irregularity and tendency to revert quickly rather than persist. Introducing the concept of roughness and offering continuous time representation aligns more with empirical evidence, suggesting irregular and anti-persistent paths.

In broad terms, the results show that models embedding roughness provide more precise estimates of future steps than the other models under consideration, especially with finer timeframes. While standard GARCH remain competitive and often perform reasonably well, it falls short in reproducing the irregular autocorrelation structure that the rough models are able to match. HAR-RV lays in between GARCH and RFSV model. Overall, the empirical analysis confirms these patterns: the RFSV achieves the lowest forecasting errors across multiple horizons, particularly in high-frequency settings, while GARCH tends to overestimate persistence and HAR-RV provides an intermediate but less accurate benchmark. These findings are consistent across different assets, reinforcing the idea that rough volatility captures essential features of market dynamics that classical approaches miss.

The results of this thesis suggest that the rough perspective can add some degree of precision in high-frequency frameworks, especially when compared to classical models. Such evidence could offer measurable improvements in risk modeling and forecasting, which could be integrated with tools in both private and public institutions.

The remainder of the thesis is organized as follows. The next section will unveil links between some stylized facts on volatility and the literature on this topics, these will bring us through the development of the idea of rough volatility in recent years. Then a modeling section where models will be explained in detail, with a particular attention to the mathematical grounding of RFSV. The Empirical Analysis section will uncover the process of applying theory to real data, with a focus on the dataset, parameter estimation and techniques for forecasting. Lastly a section in which the results will be presented and discussed in detail.

2 Stylized Facts of Volatility

In the last three decades, the study of financial volatility has progressively moved from a purely empirical recognition of recurrent phenomena to a more formal attempt to embed such regularities into models. The literature often refers to these phenomena as stylized facts, and they constitute the empirical backbone against which any proposed model of volatility should ultimately be tested. Their importance lies not only in their descriptive capacity, but also in the way they have shaped the development of increasingly sophisticated frameworks, from discrete time models to continuous time specifications inspired by stochastic calculus.

This section revisits some of the most prominent facts that have emerged in the empirical finance field, focusing in particular on those that have motivated the rise of rough volatility models. The discussion will cover important patterns such as volatility clustering, leverage effect, persistence, and roughness, as well as phenomena that are more recent in the literature, like volatility-of-volatility dynamics and microstructural influences, or the scaling laws and universal features identified across different asset classes. Each of these elements represents a piece of the puzzle, and together they provide the conceptual foundation upon which the transition from observation to formalization has been built, eventually raising questions on the interplay between memory and mean reversion. The following subsections are organized accordingly, guiding the reader from the empirical manifestations to the theoretical formulations that seek to reproduce them.

2.1 Volatility Clustering

Among the most consistently observed patterns in financial time series is the phenomenon known as volatility clustering. First noted in the context of equity markets but later confirmed across a wide range of asset classes, from fixed income to foreign exchange and commodities, it refers to the empirical tendency for large price movements to be followed by further large movements, and likewise, for small fluctuations to be followed by quiet periods. In practical terms, this means that volatility is not evenly distributed through time: instead, it tends to arrive in waves. The market will pass through turbulent regimes and calm phases, often unpredictably, and yet the fact that these regimes tend to persist, once they emerge, is remarkably robust.

What is striking about this phenomenon is that it arises even when returns themselves show little to no autocorrelation. While price changes often resemble white noise, their magnitude exhibits long stretches of persistence. In the early econometric literature, this was one of the key motivations behind models such as ARCH and GARCH, which allowed volatility to vary over time and respond to past shocks. And indeed, these models were quite successful in capturing the short term dependence structure of volatility. But their appeal often lies more in parsimony and tractability than in their ability to reflect the full

richness of observed behavior.

From a modeling perspective volatility clustering poses a fundamental challenge, it suggests that volatility is not a simple Markovian process, but one whose future trajectory depends on a broader history of past states. This is especially relevant when designing models for risk forecasting or derivative pricing, situations in which the timing of volatility bursts may matter as much as their amplitude. Furthermore, clustering behavior complicates standard assumptions in portfolio theory, where variance is simplistically often treated as constant over short horizons. Even though GARCH-type models provide a partial solution, the idea of volatility as a rough process offers a more structural interpretation, rather than reacting to shocks with decay, volatility may instead exhibit a form of irregular persistence which is not easily captured by exponential smoothing. It is precisely this distinction, between memory driven by autoregression and roughness arising from the fine structure of path behavior, that motivates much of the modern shift in volatility modeling.

2.2 Leverage Effect and Asymmetry

Another intriguing and consistently observed feature in financial time series is the so called leverage effect. It essentially points to an observed asymmetry in how market volatility responds to past price movements. Unlike a simple expectation where positive and negative returns of the same magnitude would have mirror image effects, the leverage effect suggests that negative returns tend to be followed by a disproportionately larger jump in volatility compared to the dampening effect seen after positive returns. This asymmetric reaction to news hints at deeper behavioral or structural features of markets. For example, it may reflect risk aversion, liquidity constraints, or the cascading effects of deleveraging in turbulent times. The conditional distribution of returns is not symmetric, neither in shape nor in its volatility dynamics, and any realistic model must account for this.

Traditional models have attempted to incorporate this asymmetry through various means. In discrete time settings, GJR-GARCH and EGARCH models include terms that allow for different responses to positive and negative returns. In continuous time, models like the Heston stochastic volatility framework incorporate correlation between the Brownian motions driving returns and volatility, allowing for a negative instantaneous correlation, the so called leverage parameter. While these modifications improve the empirical realism of volatility models, they often do so by introducing extra parameters rather than rethinking the structure of volatility itself.

One of the more intriguing aspects of rough volatility models is that they can naturally accommodate certain forms of asymmetry, not through additional correction terms, but as a consequence of the irregularity and memory embedded in the volatility path itself.

Though the connection between roughness and the leverage effect is still being explored, there is growing evidence that rough paths may capture both the timing and the magnitude of volatility shifts in a way that correlates more organically with return dynamics. This provides not just a technical advantage, but a conceptual shift: volatility becomes less of a reactive process and more of a structure with its own microfoundations, shaped by how information flows and risk is redistributed through the market.

2.3 Persistence and Memory

When examining the time series behavior of volatility, particularly through the lens of high-frequency data, one does not observe a pattern of isolated shocks or quickly fading influence. What appears instead is a more subtle and persistent form of dependence, where periods of high volatility tend not to vanish abruptly but to stretch into sequences, and where low volatility regimes maintain their character longer than one would expect from a memoryless or short memory framework. This persistence, which can be visualized in the slow decay of autocorrelation functions or inferred from the clustering structure of squared returns, raises a fundamental question about how financial markets transmit information and how they metabolize uncertainty across time. In many classical models, especially those built around the idea of mean reversion, such as the Heston model or similar continuous time stochastic volatility frameworks, variance is assumed to revert toward its long run mean at an exponential rate, meaning that the process forgets quickly and that past values have little effect on future behavior once some lag is reached. This assumption, while useful in practice and comforting in theory, sits uneasily with what the data tend to show.

Empirically, the decay in autocorrelation appears much slower than exponential, and this leads to the hypothesis that volatility might exhibit some memory. Models that include fractional integration or other forms of persistent structure have been proposed to capture this feature more directly. These models, at least in principle, allow shocks to fade gradually, and they provide a richer dynamic than what is permitted by standard autoregressive specifications. Some new complications arise since it is often difficult to separate the signal of true long memory from the illusions that arise when irregularity mimics dependence.

A natural benchmark when addressing persistence in volatility is the FIGARCH model. First introduced by [Baillie et al. \(1996\)](#), it generalizes the GARCH framework by allowing the conditional variance to follow a fractionally integrated process. In practice, this amounts to specifying a conditional variance equation of the form

$$(1 - \beta(L))(1 - L)^d \sigma_t^2 = \omega + [1 - \phi(L) - \beta(L)](1 - L)^d \epsilon_t^2,$$

where d is a fractional differencing parameter capturing the degree of long memory

and $\beta(L)$ denotes a polynomial in the lag operator. While this formulation is theoretically appealing, its implementation often proves cumbersome: estimation is computationally demanding, identification of d is delicate, and the finite-sample performance of estimators can be highly unstable, especially with high-frequency data.

By contrast, the classical GARCH(1,1) specification, remains far more tractable. Despite its simplicity, it was shown by [Hansen and Lunde \(2005\)](#) to provide a remarkably good approximation to volatility dynamics in many contexts, and its short memory structure makes estimation and calibration straightforward. For this reason, in empirical applications, especially with finer time scales, GARCH(1,1) often proves more practical and robust than FIGARCH, even though it does not encode long memory.

In the present work the objective is not to rely exclusively on such approximations but to move toward models that can naturally reconcile the empirical roughness of volatility with persistence effects. Rough volatility models, and in particular the RFSV, provide a framework that not only avoids the computational burdens of FIGARCH but also matches high-frequency evidence more faithfully. In this sense, they can be seen as a more powerful alternative to FIGARCH-like specifications: while the latter aims to impose long memory directly through fractional integration, RFSV generates persistence as an emergent property of path roughness. This makes it especially convenient for the analysis of high-frequency financial data, where the limitations of FIGARCH become most apparent.

Rough volatility models do not assume long memory in the classical sense, but they generate autocorrelation structures that, in many cases look remarkably similar to those produced by fractional models. The core idea is not to impose persistence through memory, but to let it emerge through the irregularity of the path itself. When volatility evolves according to a process with a Hurst exponent well below 0.5 the result is a series that is irregular at short time scales, and yet still capable of producing slow decay in standard statistical estimators of memory. This behavior complicates interpretation. As noted in the work of [Gatheral et al. \(2018\)](#) and later expanded in [Bennedsen et al. \(2022\)](#), it is often not possible to determine whether the persistence we observe in volatility is the result of true long range dependence or the illusion coming from roughness, particularly when we rely on asymptotic tools or estimation procedures that are sensitive to model specification.

For this reason, the question of memory in volatility cannot be reduced to a binary classification between short and long memory. It must instead be treated as a modeling decision, informed by the empirical regularities we observe but also by the practical constraints of estimation, calibration, and application. A model that captures persistence through roughness may be better suited for certain tasks, such as fitting the steep slope of the implied volatility surface or replicating the behavior of short term options. Another model, one that allows for explicit long memory, might offer more realism at lower frequencies or over broader time horizons. Each approach has its place, and in many

cases, the optimal strategy may involve a synthesis of the two, as recent work has begun to explore.

In the context of this thesis, the objective is not to resolve the theoretical debate but to engage with it empirically through estimation methods designed to match moments sensitive to persistence. The goal will be to examine whether a rough model, possibly augmented with additional components, can reproduce the memory observed in financial data. In this sense, memory is not only a property of the data; it is a diagnostic for model adequacy, and how we interpret it will shape the way we construct, calibrate, and eventually evaluate the models proposed in the next sections.

2.4 Vol-of-vol and Microstructure Effects

In recent years, attention has shifted from volatility itself to its own dynamics: how it fluctuates and how quickly, even whether those fluctuations carry a structure of their own. Once high-frequency realized volatility is available and volatility itself can be treated as a time series, it becomes evident that its path is not only irregular but subject to its own form of variance. Volatility has its own volatility, and this second order movement is far from negligible. We would then like to understand whether this vol-of-vol behaves according to the same principles as volatility itself, or whether it introduces new layers of complexity. Empirically, it seems that the irregularity of the volatility process is reflected and perhaps even amplified in its second order structure. Hence, if realized volatility is rough, its own variance is often rougher still.

Empirical estimates of the Hurst exponent for vol-of-vol suggest values that are even lower than those estimated for volatility. In the work of [El Euch and Rosenbaum \(2019\)](#), and in the broader literature that has followed, this has been interpreted as a sign that market uncertainty is not just time-varying, but fluctuates with high-frequency intensity that is incompatible with smooth second order dynamics. What this means in practice is that modeling volatility with a stochastic process is not enough. One must also consider the evolution of the volatility factor itself, which does not fluctuate in a purely random or irregular way, but exhibits persistent features across time scales that call for explicit modeling.

The presence of vol-of-vol becomes particularly relevant when one shifts attention from historical modeling to pricing under the risk neutral measure. In the context of derivative pricing, particularly for instruments sensitive to higher moments, the second order behavior of volatility influences skew, convexity, and the curvature of implied volatility surfaces. In rough volatility models, the presence of roughness in the vol-of-vol plays a key role in reproducing these phenomena without requiring additional jumps or regimes. El Euch and Rosenbaum formalize this through a forward variance framework that allows rough volatility models to remain tractable under the pricing measure, while still accounting for

the empirical structure of the second moment.

From a microstructural perspective this behavior is not entirely surprising. The mechanisms of modern markets, particularly those involving high-frequency trading asynchronous execution tend to produce fine scale bursts of activity that propagate into the volatility process. These mechanisms do not just create noise: they generate structure, one that becomes visible in the second order features of the data. Gatheral et al. suggest that this structure is consistent with a rough fractional regime. But it is in El Euch and Rosenbaum that we see a more concrete link between microstructure and volatility behavior, where vol-of-vol is not an abstract concept but a reflection of market mechanics acting at very short time scales.

Any attempt to model volatility realistically must confront the fact that its own fluctuations are neither smooth nor secondary. They influence pricing, estimation, and the entire dynamic of model calibration. Ignoring them leads to models that may perform well in sample but collapse when applied to real time pricing or hedging problems. Recognizing that vol-of-vol carries its own rough structure, probably induced by the microstructure of the market itself, helps to close the gap between theoretical models and empirical observations.

2.5 Roughness

The question of how volatility behaves at very short time scales has, in recent years, received increasing attention. Thanks to the availability of high-frequency data and more precise realized measures, it is now possible to observe the fine structure of volatility paths more directly than in the past. By looking closely these paths do not appear smooth, nor do they resemble the trajectories implied by classical stochastic volatility models. The variability does not dampen as the sampling frequency increases, and the process shows no signs of convergence toward a "differentiable" curve. Instead, volatility remains irregular with fluctuations that persist even when examined over very short intervals.

To formalize this idea, researchers have turned to the Hurst exponent, a parameter that describes the roughness or smoothness of a stochastic process. In models based on Brownian motion, the exponent is equal to $H = 0.5$, corresponding to a path that is continuous but nowhere differentiable. However, when one estimates H for realized volatility, the values are significantly lower. Studies such as those by [Gatheral et al. \(2018\)](#) and [Bayer et al. \(2016\)](#) report estimates around $H = 0.1$, with some variation depending on the methodology and the type of data used. These numbers point to a structural property of the process, one that seems to persist across time periods, instruments, and markets. Apparently, the low regularity of volatility paths is not an artifact of microstructure noise.

This has led to the idea that volatility should be modeled as a rough process, where the term rough refers not to any intuitive notion of unpredictability, but to a precise

mathematical property: low Hölder regularity. In simple terms, a function f is said to be Hölder continuous of order α if there exists a constant C such that for all s, t

$$|f(t) - f(s)| \leq C|t - s|^\alpha.$$

The higher the exponent α , the smoother the path. Having low Hölder regularity means that the process fails to satisfy this condition for any exponent larger than the Hurst parameter H , so when $H < 0.5$ the paths oscillate so violently at fine scales that they are rougher than Brownian motion. In this framework, volatility evolves as a path that is rougher than Brownian motion, with sample trajectories that fluctuate intensely even at high resolution. What makes this particularly relevant for modeling is the fact that many phenomena in financial markets (such as the steep slope of the implied volatility skew for short maturities) are difficult to reproduce using classical models, but emerge naturally when the underlying volatility process is assumed to be rough.

Roughness is not introduced as an additional feature layered onto an existing structure, it is embedded in the way the process is defined. Models such as the rough Heston model do not require additional sources of randomness or switching regimes. They rely instead on a different temporal structure, one that is capable of capturing the irregular, persistent behavior observed in realized volatility data. This approach does not seek to increase model flexibility by introducing more parameters. It changes the underlying assumption about how volatility behaves through time.

Understanding this difference is important when interpreting what volatility models are actually doing. A model that assumes smoothness where the data shows irregularity will fail to reproduce important features of derivative prices, particularly those that are sensitive to short term fluctuations. At the same time, adopting a rough structure allows the model to remain parsimonious, while improving its fit to observable quantities such as forward variance curves or implied volatilities near the origin. What matters here is not the complexity of the model, but its ability to reflect the empirical evidence in a way that is both precise and consistent with how volatility behaves at the smallest scales.

2.6 Scaling Laws and Universality

One of the more robust empirical findings is the fact that many of the statistical features of volatility appear to replicate themselves across different markets, instruments, and timeframes. The clustering of volatility, the asymmetry in its response to returns, the persistence of variance over time, and most notably its roughness, all of these properties are not specific to a given asset class or to a particular period of time. These features rather seem to reappear, with slight variations in scale or intensity, across equity indices, individual stocks, foreign exchange rates. This recurrence invites the question of whether there exists, underneath market-specific microstructure and institutional fric-

tions, a shared statistical structure in how volatility behaves.

Scaling laws offer one way to formalize this intuition. When one analyzes volatility at multiple time horizons (like RV for different frequencies) the patterns in variance persistence, the decay of autocorrelations, and even the estimated regularity of the paths tend to follow stable relationships. In many cases, the scaling of volatility with respect to time follows power-law behavior. These relationships are not guaranteed by the assumptions of standard stochastic volatility models, which often rely on exponential decay or memoryless transitions, instead, they suggest that volatility carries a temporal structure that is self-similar over nested time intervals, even if that self-similarity is only approximate.

Empirically, this scaling behavior reinforces the idea that volatility should be understood not as a local property of returns, nor as a regime specific artifact, but as a process governed by more general principles. When the same roughness exponent can be estimated on a currency pair and on a major equity index, or when similar autocorrelation structures appear in different geographies or regulatory regimes, it becomes difficult to attribute these properties to market design or institutional features alone. It is more plausible that the rough structure observed in volatility reflects something fundamental in how price formation occurs, particularly in the presence of heterogeneous market operators, order splitting, and asynchronous information processing.

This universality has important implications for modeling. A model that successfully captures volatility dynamics in one asset class should be transferable to other contexts if grounded in these shared statistical features. This is one of the motivations behind adopting rough volatility frameworks, which do not rely on market specific assumptions or calibration methods, but begin instead from observed regularities in the data. The fact that these regularities tend to emerge consistently lends credibility to the idea that volatility has a common geometry, one that a successful model should be able to replicate.

Still, careful interpretation is required. The existence of scaling laws and apparent universality does not imply that all markets behave identically, nor that a single model can fit every dataset without adjustment. What it suggests is that beneath this noise and heterogeneity there are structural features of volatility that hold across a wide range of empirical settings. Any model that fails to account for these features may be adequate in one narrow domain, but it will struggle to generalize. For a modeling approach to be considered robust, especially in the perspective of cross-asset applications, it must align not only with theory or intuition, but with this broader empirical stability observed in the data.

2.7 On Memory and Mean Reversion

Once the empirical features of volatility have been presented in depth, the next step is to ask what type of process could produce such behavior. In doing so one quickly encounters

a fundamental ambiguity, or rather a conceptual fork: should we interpret the slow decay of autocorrelations as evidence of long memory, or is it instead a statistical effect that emerges from pathwise irregularity, without requiring true long range dependence? And if volatility is mean reverting, as most economic intuition would suggest, what does this mean in the presence of rough paths, or in models that lack a clear stationary structure?

Traditionally, the distinction has been framed in terms of the autocorrelation function. Long memory processes are defined by autocorrelations that decay at a hyperbolic rate, meaning that their integral diverges, and their influence persists over arbitrary time horizons as noted by [Bennedsen et al. \(2022\)](#).

$$\rho(k) \sim L(k) k^{-(1-2H)} \quad \text{as } k \rightarrow \infty, \quad \text{with } H > 0.5. \quad (1)$$

Models built on this foundation often involve fractional integration, or at least autoregressive structures with coefficients that decay slowly. These models have been used to explain the persistence of volatility over weeks or months, and in some contexts, they produce realistic volatility surfaces or accurate risk forecasts. But they also bring with them a set of complications: parameter estimation becomes fragile, stationarity is often lost, and simulation or inference requires more structure than is usually available from market data alone.

The rough volatility literature proposes a different explanation. Rather than assuming persistent dependence over long time horizons, it argues that the slow decay observed in volatility is a byproduct of how the process behaves at very short scales. Specifically, models based on fractional Brownian motion with $H < 0.5$ produce sample paths that are highly irregular, yet do not exhibit long memory in the formal sense. What they do exhibit, however, is apparent persistence, especially when empirical estimators (such as Realized Volatility) are applied to them. This is because roughness, while mathematically short memory, still produces a kind of practical stickiness in the behavior of the process, which can resemble long memory when viewed at the rougher scales used in real data.

$$\rho(k) \sim L(k) k^{-(1-2H)} \quad \text{as } k \rightarrow \infty, \quad \text{with } H < 0.5. \quad (2)$$

A more recent contribution, due to [Bennedsen et al. \(2022\)](#), proposes a modeling approach that attempts to disentangle the effects of short scale irregularity from those of long range dependence, without assuming that one excludes the other. In their formulation, volatility is allowed to fluctuate with a high degree of local roughness while at the same time preserving a slower form of persistence that unfolds across broader time horizons. The implication is not that volatility must be classified as either short memory or long memory, but that both behaviors may be present, layered within the same process and operating at different temporal resolutions. Roughness, in this view, dominates at the small scale, where the path exhibits sharp fluctuations and low regularity,

while persistent components may emerge more clearly when volatility is aggregated or averaged over longer windows. This perspective does not resolve the question of which feature is more fundamental, but it does shift the focus: rather than treating memory as a binary property, it becomes something to be understood in terms of scale, structure, and empirical scope.

What this means for modeling is that we must be cautious in how we interpret empirical evidence. A power-law decay in autocorrelations does not imply long memory on its own, nor does it rule it out. The presence of short memory dynamics in a model does not guarantee that it will fail to reproduce persistence in volatility, especially if the process is rough. The modeling choice hence becomes less about fitting a specific decay pattern, and more about understanding which properties matter most for the intended application. If one is interested in capturing short term risk, or pricing short dated options, roughness may be the dominant feature. If the focus is on long run forecasts then persistence may still require explicit modeling in some cases.

This section does not resolve the tension between memory and mean reversion, nor does it advocate for one structure over another. Still, it highlights the fact that this tension is structural and that models which attempt to reproduce empirical features must confront it directly. The challenge is to build frameworks that are not only able to mimic observed behaviors, but also precise enough in their assumptions to be interpreted meaningfully. And this is particularly true when memory, mean reversion, and roughness are all in play, each suggesting a different picture of what volatility is, and what it remembers.

3 Modeling Volatility

Before one attempts to formalize any stochastic model of volatility, it is crucial to reflect on what a good model in this field really is. We are trying to see if certain models can reproduce what we observe in the real world, under realistic conditions and without leaning too heavily on assumptions that only work in theory. Hence, the modeling of volatility is not just parameterizing but rather interpreting: a volatility model must capture the empirical features that persist across markets and time, while remaining tractable enough to be estimated and tested with real data. In this section, we shift from the description of the question to the actual implementation, moving carefully from the stylized facts described earlier to a set of formal, estimable models that aim to replicate them under the historical probability measure $\mathbb{P}$. The objective is not to simulate volatility in the abstract, but to construct and calibrate processes whose behavior mirrors the temporal and statistical structure of observed market volatility. To evaluate and compare models of volatility in a meaningful way, we must first clarify the criteria by which such models are judged. In the context of this thesis, a model does not succeed by maximizing likelihood, but by reproducing certain core properties of realized log-volatility as it evolves under $\mathbb{P}$. These properties are not arbitrary: they reflect robust empirical patterns that have been described in literature, and they include long-range dependence, clustering behavior, and a specific kind of pathwise irregularity often referred to as "roughness". Concretely, we say that a model fits volatility if it is able to generate simulated paths of log-volatility whose statistical features (on average and across simulation replications) align closely with those found in the data. This alignment is assessed through the lens of moment-based diagnostics: the autocorrelation function, the scaling behavior of aggregated variances, and possibly higher-order moments such as skewness or kurtosis. At the heart of this evaluation lies the question of persistence: how far into the future does the current level of volatility influence the process? How does this influence decay? A good model, in this case, is one that produces the right kind of memory, which is not an infinite one, but with enough persistence to mimic the hyperbolic decay seen in empirical autocorrelations.

But there is more: beyond just memory, a model must replicate the roughness of volatility paths, meaning their tendency to flicker even at very short time scales. This is not a mere nuisance or an artifact of high-frequency noise, but a defining feature of volatility itself, as discussed in the previous sections. Therefore, our notion of model fit includes not only statistical fidelity in moments, but also structural compatibility with the fine-scale geometry of the process. This property is not merely an artifact of high-frequency noise, but a deep defining feature of the volatility series, which we will not search through fit in moments, but rather by structural comparison.

In what follows, we describe three models: a GARCH(1,1), a HAR-RV, and a fractional rough volatility model, each estimated and simulated in a way that allows for direct

empirical comparison on this basis. Once a model has been specified and its parameters estimated, then it is time to judge how it performs, i.e., how well does it resemble the observed behavior of volatility. As previously mentioned in the Data section, we did not embrace the current dominant field of analysis represented by high frequency options data, but decided to go deeper into the realm of $\mathbb{P}$, using Standard and Poor's 500 as the most representative index for the whole world equities.

3.1 GARCH model

The GARCH (Generalized Auto Regressive Conditional Heteroskedasticity) model, introduced by [Bollerslev \(1987\)](#) and [Engle and Bollerslev \(1986\)](#), is a classic approach to volatility modeling that captures the well-known phenomenon of volatility clustering. In a GARCH model, the conditional variance at time t (denoted σ_t^2) depends on its own past value and on the most recent squared return innovation. We define return innovations as

$$\epsilon_t = r_t - \mu_t, \quad (3)$$

where r_t denotes the observed return at time t and $\mu_t = \mathbb{E}[r_t \mid \mathcal{F}_{t-1}]$ is its conditional mean given the information set $\mathcal{F}_{t-1}$. The standard GARCH(1,1) specification for the conditional variance is then

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (4)$$

with $\omega, \alpha, \beta \geq 0$ and typically $\alpha + \beta < 1$ to ensure covariance stationarity. Intuitively, α controls the short-term reaction of volatility to market shocks (also known as "ARCH effect"), while β governs the persistence or slow decay of volatility following a shock (the so called "GARCH effect"). Despite its parsimony, the GARCH(1,1) model can generate a decaying autocorrelation in volatility and thereby reproduce the clustering of large moves followed by periods of relative calm. However, the decay of volatility autocorrelations in a GARCH model is exponential (with rate $\alpha + \beta$) rather than hyperbolic, this implies that unlike long memory processes, GARCH does not capture extremely long range dependence if not specified differently.

GARCH(1,1) parameters were estimated by maximizing the likelihood of the observed returns, assuming a normal error distribution for ϵ_t .

$$\ell_t(\varphi) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} z_t(\varphi)^2 - \frac{1}{2} \log(\sigma_t^2). \quad (5)$$

Maximizing the sum $\ell(\varphi) = \sum_{t=1}^T \ell_t(\varphi)$ with respect to $\varphi = (\omega, \alpha, \beta)$ yields the maximum likelihood estimates of the model. This likelihood-based estimation ensures statistical efficiency under correct specification of the innovation distribution. In this application this

was combined with a rolling-window scheme, instead of fixing the parameters over the whole sample, we re-estimate them on a moving window of past observations. This choice is motivated by the need to continuously incorporate new information while preventing outdated observations from exerting undue influence on the estimation. Financial volatility is known to exhibit structural changes and regime shifts. Thus, by re-estimating the model on a rolling window of recent data, the GARCH parameters remain aligned with current market conditions, hence improving the relevance and accuracy of short-term forecasts. Such approach allows GARCH parameters to adapt over time, reflecting structural changes in market volatility. Since GARCH has a short memory, its forecasts beyond a few periods ahead tend to revert quickly to the unconditional variance level given by $\omega/(1 - \alpha - \beta)$, this is the reason behind the rolling window approach in this thesis.

3.2 HAR-RV model

Realized volatility has become the standard proxy for the latent variance of asset returns, since high-frequency data allow to estimate the integrated volatility of a price process with considerable accuracy. Given intraday returns $r_{t,i}$ over a trading day t , sampled at a sufficiently fine grid, realized variance is defined as

$$RV_t = \sum_{i=1}^M r_{t,i}^2, \quad (6)$$

where M denotes the number of intraday intervals. The square root of RV_t is commonly referred to as realized volatility, and it is widely employed as an ex-post measure of daily variability. This construction is appealing because it provides a nonparametric estimate of the quadratic variation of returns, while at the same time circumventing the unobservable nature of conditional variance in traditional GARCH-type models as in [Bennedsen et al. \(2022\)](#); [El Euch and Rosenbaum \(2018\)](#).

The Heterogeneous Autoregressive Realized Volatility model (HAR-RV), introduced by [Corsi \(2009\)](#), was designed to capture the strong persistence observed in realized volatility series in a simple, additive manner. The HAR-RV model acknowledges that different market participants or factors operate on different time scales (daily, weekly, monthly), and thus it incorporates volatility components over multiple horizons. The standard HAR-RV model specification can be expressed as:

$$RV_{t+1} = \beta_0 + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \epsilon_{t+1}, \quad (7)$$

where $RV_t^{(d)}$ is the realized volatility on a trading day t , $RV_t^{(w)}$ is the average daily realized volatility over the past trading week (i.e., the mean of the last 5 daily values up to day t), and $RV_t^{(m)}$ is the average daily realized volatility over the past trading month (i.e.,

approximately the last 22 trading days up to day t). The coefficients $\beta_d, \beta_w, \beta_m$ capture the influence of short-term, intermediate-term, and longer-term volatility factors. A key insight of the HAR-RV model is that by combining these heterogeneous durations, it can approximate the behavior of a long-memory process within a simple linear regression framework. In particular, if β_w and β_m are sizable, a shock to volatility will have a prolonged influence (through the weekly and monthly terms), much longer than it would in a standard AR(1) model.

The estimation of the HAR-RV model is straightforward, since equation (7) is linear in the parameters. Given the realized volatility series, one constructs the lagged regressors $RV_t^{(d)}$, $RV_t^{(w)}$, and $RV_t^{(m)}$ and estimates the coefficients $\beta_0, \beta_d, \beta_w, \beta_m$ by ordinary least squares (OLS). In this implementation the model is applied to the log RV series (using the logarithm of realized variance) to mitigate skewness and heteroskedasticity. As in the case of GARCH estimation is carried out with a rolling window approach, for each forecast origin the regression is re-fitted on the most recent chunk of data, then the one-step-ahead forecast is obtained by plugging the current lag values into (7). This approach ensures that the HAR-RV model remains aligned with the current environment, it avoids overweighting outdated volatility regimes, and yields forecasts that are genuinely out-of-sample, as they are based exclusively on the information set available up to time t .

3.3 RFSV model

As developed by [Comte and Renault \(1998\)](#) the Fractional Stochastic Volatility model for the first time introduced the concept of fractional stochastic process to the volatility modeling framework. In that paper they proposed to model log-volatility using a fractional Brownian Motion process but they ensured to keep long memory as a property by having $H > 0.5$. This approach was then further developed in the 2000's by other scholars, all of them pursued the "long memory" venue of research with good results. In 2014 a pioneering research by [Gatheral et al. \(2018\)](#) introduced the Rough Fractional Stochastic Volatility (RFSV) model, where $H < 0.5$ and was estimated to lie in between 0.08 – 0.2, which was interesting because we see that this approach, not only introduces roughness to the process, but it ensures long memory even though this property is not explicitly imposed. Here RFSV will be introduced formally through the methodology that was used to estimate it and make predictions.

The model assumes that the log-volatility process, $\log \sigma(t)$, follows a scaled Fractional Brownian Motion:

$$\log \sigma(t) = \nu W^H(t) , \quad (8)$$

where $W(t)$ is a Fractional Brownian Motion with Hurst parameter H , and ν is a scaling factor for volatility of volatility. This specification implies that increments of $\log \sigma(t)$ follow fractional Gaussian noise with parameter H , which governs both the roughness

and the memory characteristics of the volatility path.

In a rough setting the Hurst parameter is usually measured to be in this interval:

$$H \in (0, \frac{1}{2}) .$$

While fBm is not a semimartingale and thus cannot drive asset prices directly in the classical Itô framework, it has proven useful in modeling latent components such as volatility. In fact, several models developed under the risk-neutral measure $\mathbb{Q}$, such as the rough-Bergomi model, postulate a volatility process of the form:

$$\log \sigma_t = \eta \int_0^t (t-s)^{H-1/2} dW_s^{\mathbb{Q}}, \quad (9)$$

where $W_s^{\mathbb{Q}}$ is a Brownian motion under $\mathbb{Q}$, and $H < 0.5$ captures the roughness. Equation (9) is just one example of models coming from the risk-neutral measure literature, since most research comes from the derivatives pricing application.

Inspired by this structure, and supported by strong empirical evidence [Gatheral et al. \(2018\)](#), we postulate under the historical measure $\mathbb{P}$ a direct model for log-volatility as in (8), where $W_H(t)$ is a fractional Brownian motion with Hurst exponent $H \in (0, 0.5)$. This model is not derived from a no-arbitrage pricing argument, but rather designed to replicate the observed statistical features of volatility paths under $\mathbb{P}$.

It is pivotal to this discussion how the RFSV model can capture roughness and long memory at the same time ([Mandelbrot and Van Ness \(1968\)](#)). Traditional long-memory processes (with $H > 0.5$) are stationary and have autocorrelation functions that decay hyperbolically with lags. In this case, $\log \sigma(t)$ as defined in (8) is non-stationary in level, so we instead consider the behavior of its increments, which form a stationary fractional Gaussian noise (fGn). For an fBM with $H < 0.5$, the autocorrelation of increments at lag k is:

$$\rho(k) = \frac{1}{2} \left((k+1)^{2H} - 2k^{2H} + (k-1)^{2H} \right), \quad (10)$$

which for large k behaves as:

$$\rho(k) \sim H(2H-1) k^{2H-2}. \quad (11)$$

Another interesting property of the RFSV model is its behavior across different sampling frequencies. Many classical volatility models (including GARCH and HAR-RV) are not invariant to the frequency of observation. Their parameters and fit can change significantly if we calibrate them to daily data versus hourly or 5-minute data. This happens because of mainly two reasons: microstructure noise at very high frequencies and because such models lack a self-similarity feature built in the model specification. The RFSV model is instead built on a self-similar process: if $\log \sigma(t)$ follows (8) with a given

H , then aggregating or sub-sampling the process does not change the value of H significantly. The scaling law (13) implies that the Hurst exponent is an intrinsic property of the volatility path, not an artifact of the sampling interval. Therefore, as a diagnostic, we can estimate $\hat{H}$ on volatility data sampled at different frequencies and check for consistency. In our analysis, we do exactly this, applying the estimation method to realized volatility computed at 5-minute, hourly, and daily frequencies. The expectation is that the estimates of H will be roughly the same in all cases. Any large discrepancies that we may have observed would have indicated that either model misspecification or significant impacts of microstructure noise (or both) at the highest frequency happened. This scale-invariance of RFSV is a major conceptual advantage, as it suggests the model is capturing a fundamental aspect of volatility that prevails across all timescales.

A crucial step when dealing with this model is estimating the Hurst exponent H from data. We employ a *log-log regression* approach based on the scaling properties of fractional Brownian motion. The idea is that for an fBM, the q -th absolute moment of increments scales as a power of the time increment:

$$\mathbb{E}\left[|W_H(t+\tau) - W_H(t)|^q\right] = C_q \tau^{qH}, \quad (12)$$

for some constant C_q that depends on q . Taking logs of both sides yields a linear relationship:

$$\log \mathbb{E}\left[|W_H(t+\tau) - W_H(t)|^q\right] = qH \log(\tau) + \log C_q, \quad (13)$$

which suggests that if we compute empirical q -th moments of increments of $\log \sigma(t)$ at various time lags τ , the plot of $\log(\text{moment})$ versus $\log(\tau)$ should approximately be linear. In practice we chose a moment order ($q = 2$ for variance) and estimate H as follows: for a range of lags (i.e., 1 day, 5 days, 10 days or more), compute the sample moment:

$$m_q(\tau) = \frac{1}{N - \tau} \sum_{t=\tau+1}^N |\log \sigma(t) - \log \sigma(t - \tau)|^q, \quad (14)$$

Then we perform a linear regression of $\log m_q(\tau)$ on $\log \tau$. The β of this regression (slope) provides an estimate of qH , and thus we can set $\hat{H} = \beta/q$.

$$m_q(\tau) = \frac{1}{N - \tau} \sum_{t=\tau+1}^N |X_t - X_{t-\tau}|^q \quad (15)$$

By plotting the logarithm of these sample moments against the logarithm of the corresponding time lags, we obtain the linear relationship:

$$\underbrace{\log(m_q(\tau))}_y = \beta \underbrace{\log(\tau)}_x + \alpha, \quad (16)$$

hence, the slope β provides an empirical estimate of qH , from which the final Hurst estimate is:

$$\hat{H} = \frac{\beta}{q}. \quad (17)$$

The intercept of the regression relates to $\log C_q$ and can be used to determine the scale factor ν . It is possible to show that, for $q = 2$, $\nu^2 \approx C_2$ if $\log \sigma(t)$ exactly followed (8)).

Once we have estimates $\hat{H}$ and $\hat{\nu}$, we proceed to leverage the Gaussian nature of the fBM to forecast the volatility. The log-volatility path (past and future) is jointly Gaussian with mean zero and a covariance structure determined by H and ν . In particular, the covariance between any two times t_i and t_j is:

$$\mathbb{E}[W_H(t_i) W_H(t_j)] = \frac{1}{2} \left(t_i^{2H} + t_j^{2H} - |t_i - t_j|^{2H} \right), \quad (18)$$

which we use to construct the covariance matrix Σ for all needed time points.

Suppose we have observed log-volatility up to time t_N and wish to forecast h steps ahead. We partition the full series into two segments: the past $W_p = [W_H(t_1), \dots, W_H(t_{n_p})]^T$ and the future $W_f = [W_H(t_{n_p+1}), \dots, W_H(t_{n_p+h})]^T$. The full process is modeled as:

$$\begin{pmatrix} W_p \\ W_f \end{pmatrix} \sim \mathcal{N} \left(\mathbf{0}, \Sigma = \begin{pmatrix} \Sigma_{pp} & \Sigma_{pf} \\ \Sigma_{fp} & \Sigma_{ff} \end{pmatrix} \right) \quad (19)$$

From the properties of conditional multivariate normals, the predictive distribution for f given p is Gaussian with mean:

$$\mu_{f|p} = \Sigma_{fp} \Sigma_{pp}^{-1} p, \quad (20)$$

and covariance:

$$\Sigma_{f|p} = \Sigma_{ff} - \Sigma_{fp} \Sigma_{pp}^{-1} \Sigma_{pf}.$$

The conditional mean vector $\mu_{f|p}$ provides the best (i.e., minimum loss) forecast for the future log-volatility values. We computed this section by solving the linear system in (20), practically regressing the future on the past in a Gaussian setting. The resulting forecast for $\log \sigma(t)$ over the next h steps is then given by $\log \hat{\sigma}_f = \nu \mu_{f|p}$, because $\log \sigma(t)$ is defined as ν times $W_H(t)$. Confidence bands or sample future scenarios can be thus generated by drawing from $\mathcal{N}(\mu_{f|p}, \Sigma_{f|p})$.

When $H < 0.5$, the coefficient $H(2H - 1)$ is negative. This implies that long-lag increments are negatively correlated, which is the anti-persistence property previously introduced. Moreover, $2H - 2 < -1$, so $\sum_{k=1}^{\infty} \rho(k)$ converges (in the strict sense), the process does not have infinite memory as shown by Gatheral et al. (2018). The autocorrelation decay k^{2H-2} is still hyperbolic and very slow for H in the rough volatility range we are working with ($H \in [0.08 - 0.2]$). This implies that volatility shocks have effects that

persist over extremely long horizons, and then decay out. We see that RFSV model captures the empirical reality of volatility. Even though its increments may be anti-persistent at the very high frequency level, there remains a form of long range dependence evident in how volatility fluctuations decay over time. The model thus provides an explanation for why volatility can appear to have long memory despite the underlying process being much rougher than classical models would assume.

4 Empirical Analysis

This section examines how the theoretical framework is applied to real-world constraints. The implementation follows a pipeline that allows for the comparison of GARCH, HAR-RV and RFSV models across multiple frequencies and for a broad set of assets, exploiting both historical returns and realized measures of volatility.

The first subsection describes the data, covering the construction of realized volatility from high frequency observations, the preparation of lag structures necessary for HAR estimation, and the transformation of returns for GARCH specification. In this stage, particular attention is devoted to the alignment of daily, hourly and five-minute series, as well as to statistical diagnostics such as the Augmented Dickey–Fuller tests, which provide a preliminary assessment of persistence and stationarity in the volatility process.

The second subsection presents the parameter estimates obtained through rolling window procedures. For GARCH models, this involves the recursive estimation of (ω, α, β) coefficients on blocks of returns, ensuring stability and continuity across forecast horizons. HAR-RV parameters are estimated via ordinary least squares within rolling windows, capturing the evolving contribution of daily, weekly and monthly components to realized variance. In the case of RFSV, the procedure is more intricate, relying on log–log regressions of absolute increments to identify the Hurst exponent H and the volatility scale parameter ν , before generating forecasts based on the conditional expectation of fractional Brownian motion increments. These estimation routines, while different in their econometric logic, are designed to be structurally comparable in order to highlight the contrasting dynamics each model emphasizes.

Finally, the third subsection deals with in-sample and out-of-sample analysis, where model forecasts are evaluated against realized volatility over holdout periods of fixed length. This part of the study employs a battery of metrics, from classical root mean squared error to more distributional criteria such as the continuous ranked probability score, allowing a multi-dimensional assessment of forecast accuracy. The evaluation is carried out both at the aggregate level, by summarizing performance across all tickers and frequencies, and at the individual level, where results are inspected ticker by ticker to uncover possible heterogeneity in model behavior. Visualizations in the form of heatmaps, comparative plots and forecast trajectories complement the statistical results, providing an intuitive representation of strengths and weaknesses.

This section provides the empirical counterpart to the theoretical discussion, illustrating in practice how the different specifications adapt to the data and to what extent the rough paradigm can be justified in comparison with classical econometric approaches.

4.1 Data

In the modeling framework, the decision on the type of data and their frequency becomes crucial. These decisions depend on which way we decide to tackle the research question and we take a look at the literature for potential guidance in the whole realm of possibilities. There are mainly two ways of addressing a solution to this research question: using option prices under $\mathbb{Q}$ (as in [Gatheral et al. \(2018\)](#) and [El Euch and Rosenbaum \(2019\)](#)) or choosing to opt for $\mathbb{P}$ which represents the underlying price dynamic. First, as argued in [Bennedsen et al. \(2022\)](#), volatility under $\mathbb{P}$ exhibits both roughness and persistence, which are stylized features visible across a broad spectrum of assets, time periods, and asset classes, and that emerge irrespective of any risk-neutral adjustment. In this sense, roughness is not merely a mathematical artifact tuned to fit the implied volatility surface, but an intrinsic statistical property of volatility itself. The authors show that even at low frequencies, volatility processes display the non-Markovian, fractal-like behavior that motivates rough modeling, suggesting that this is not a feature isolated to short-term option pricing models, but a deeper structural characteristic of financial time series.

Second, by avoiding $\mathbb{Q}$, we remove the dependence on option prices—which are contaminated by liquidity effects, market frictions, and time-varying risk premia that are really hard to estimate robustly. As noted in [Gatheral](#), forward variance curves under $\mathbb{Q}$ depend intricately on both historical volatilities and the market price of risk, which in turn introduces ambiguity and instability in estimation. In contrast, modeling under $\mathbb{P}$ allows us to work with directly observed series of returns and realized variances, with minimal assumptions about investor preferences or equilibrium. The most important single aspect that led us to this choice is that forecasting is inherently a task under $\mathbb{P}$. When we build volatility models for the purpose of risk management or portfolio allocation we are interested in how volatility evolves in real time, not how it is priced by options markets. Our commitment to evaluating predictive performance directly under $\mathbb{P}$ reflects this practical orientation. Hence, we explicitly want to discover if using rough models under $\mathbb{P}$ it is possible to predict volatility better than using the dominant historical benchmarks.

After motivating our choice of using $\mathbb{P}$ over $\mathbb{Q}$ we would like to introduce the price series of interest: we analyze a panel of nine of the most liquid stocks in the global market and the Standard and Poor’s 500, as in later Figures. This diversified selection allows us to test the robustness of our volatility modeling framework across different sectors and asset profiles. Each of these assets is widely traded and reflects key segments of financial markets.

The time window spans from July 2023 up until July 2025 for the daily frequency, 60 trading days between April 2025 and July 2025 for hourly and 5-minutes frequency. This timespan was selected according to two competing needs: stationarity and representative-

ness of the series. Ideally we would not want to include really big exogenous shocks (such as the 2020 Covid pandemic), hence we were able to dodge the post-pandemic transition regime, where volatility was highly irregular and financial markets were still digesting the aftermath of the global crisis. At the same time including the start of 2025 ensures that the sample captures a full cycle of volatility conditions, mostly because of the spikes due to the Tariffs war initiated by Trump that split market sentiment.

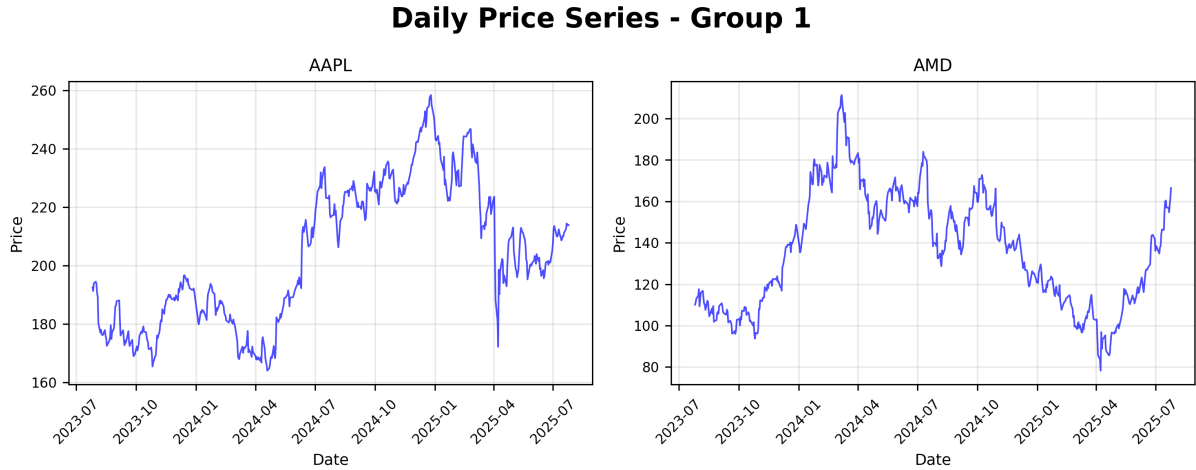


Figure 1: Evolution of Apple and AMD price over time for daily frequency

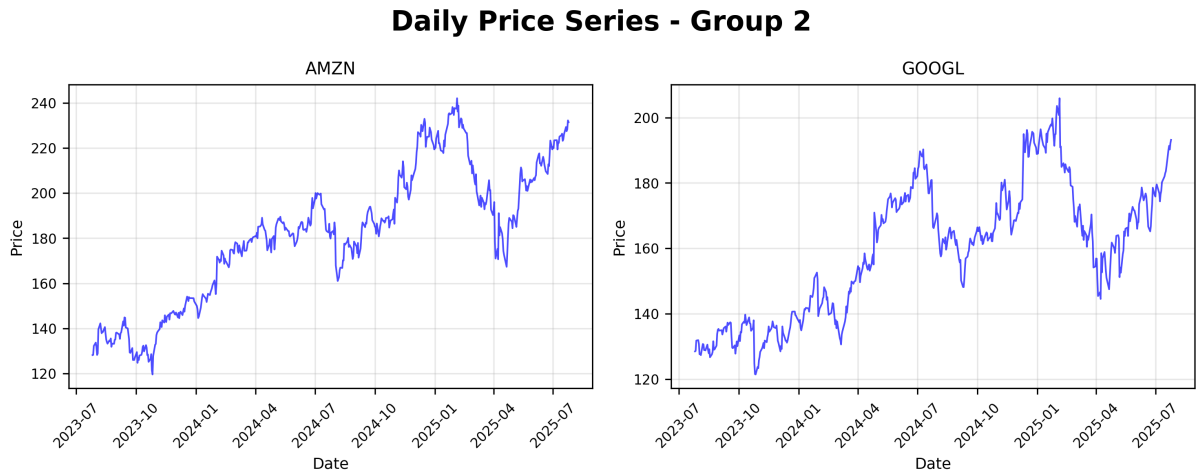


Figure 2: Evolution of Amazon and Google price over time for daily frequency

This window provides a sufficiently rich environment to test whether models driven by fractional Brownian motion, without mean reversion or exogenous regime switches, can account for the endogenous fluctuations in volatility observed under $\mathbb{P}$. By iterating model estimations over daily, 1-hour and 5-minutes frequencies it was possible for our model to be compared with the institutional forecasting ones, which is the latter part of our research proposal.

This dataset can be freely downloaded online from my Github repository and contains data on multiple assets at a 5-minute frequency. Originally these data were scraped

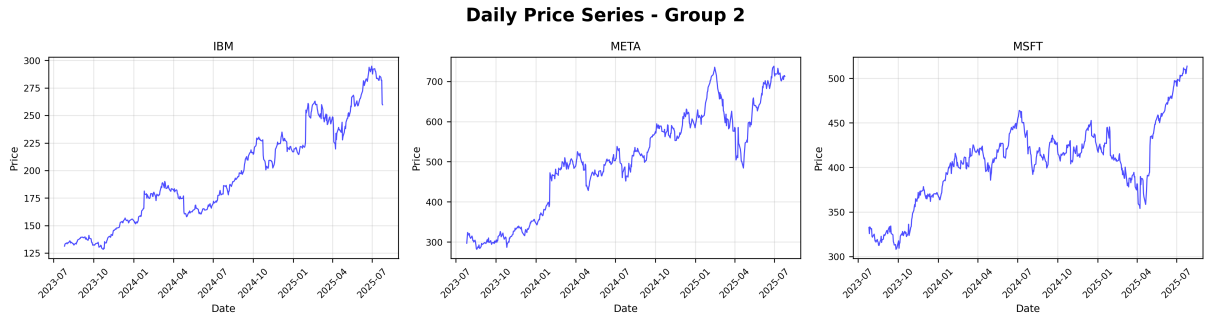


Figure 3: Evolution of IBM, Meta and Microsoft price over time for daily frequency

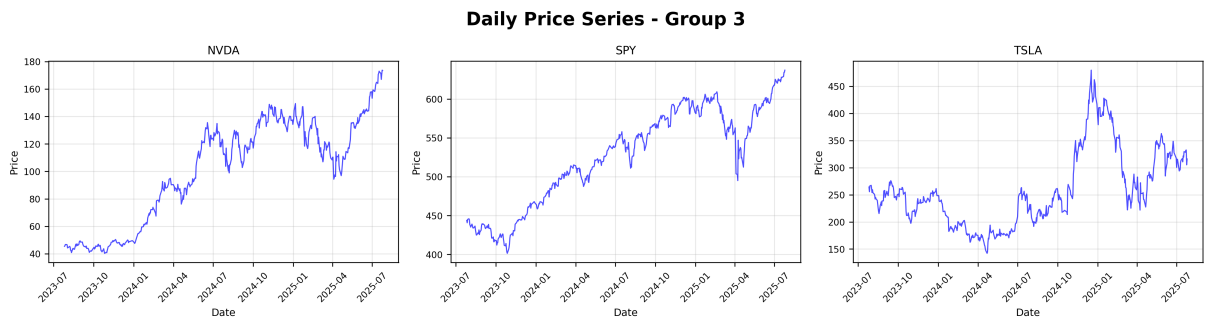


Figure 4: Evolution of Nvidia, Standard and Poor's 500 and Tesla price over time for daily frequency

through Yahoo Finance library and then stored in this Github repository. Another argument in favor of working under $\mathbb{P}$ instead of $\mathbb{Q}$ is that data on the first is usually available freely online, even at high frequency, whereas the second one is beyond paywall, especially for long horizons.

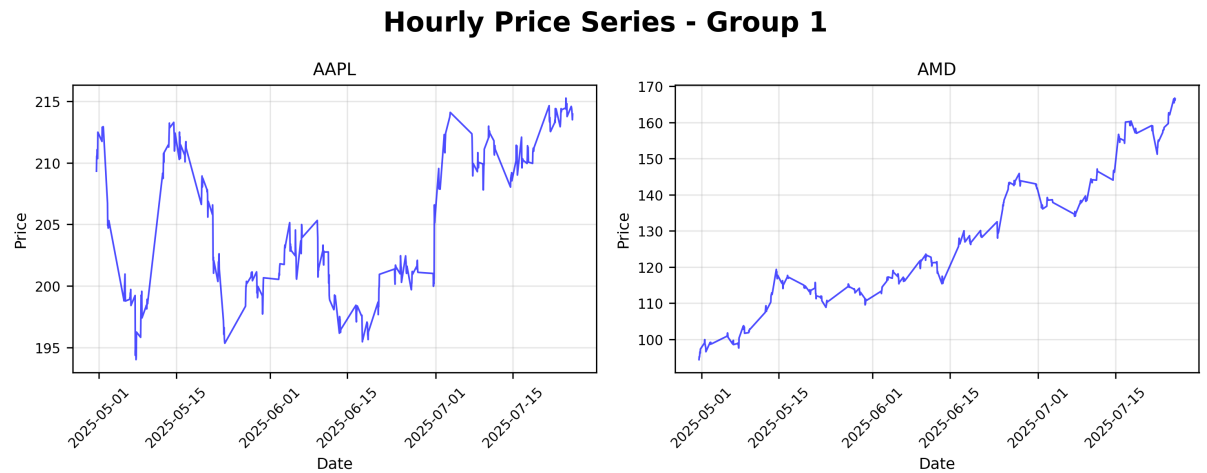


Figure 5: Evolution of Apple and AMD price over time for hourly frequency

Realized variance from log-returns is our variable of interest. This transformation is not merely a matter of numerical convenience but reflects a modeling stance. It is well established in financial econometrics that log-return is the most convenient measure be-

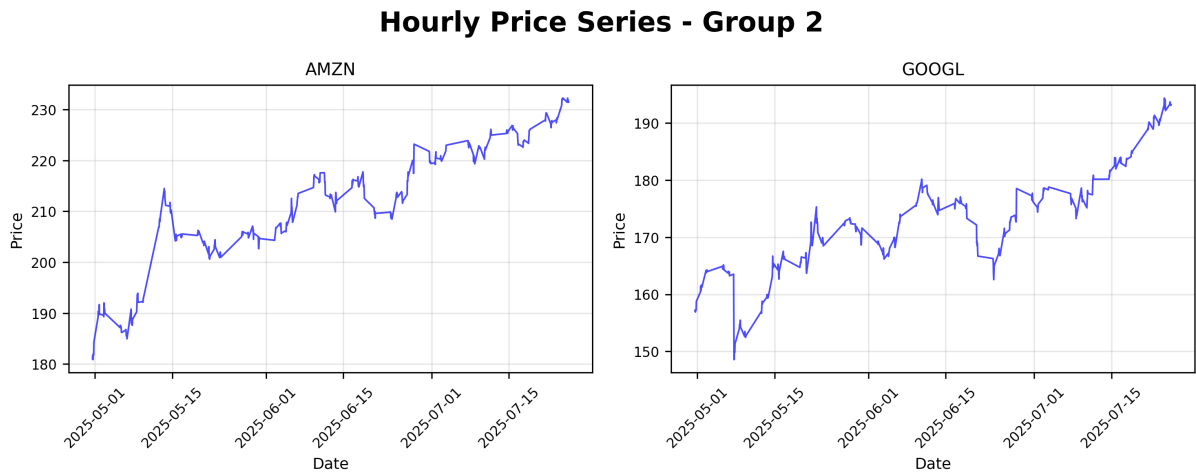


Figure 6: Evolution of Amazon and Google price over time for hourly frequency

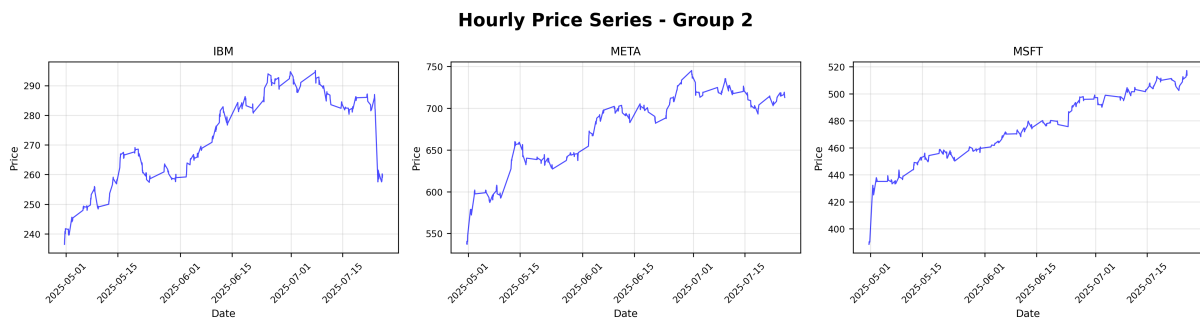


Figure 7: Evolution of IBM, Meta and Microsoft price over time for hourly frequency

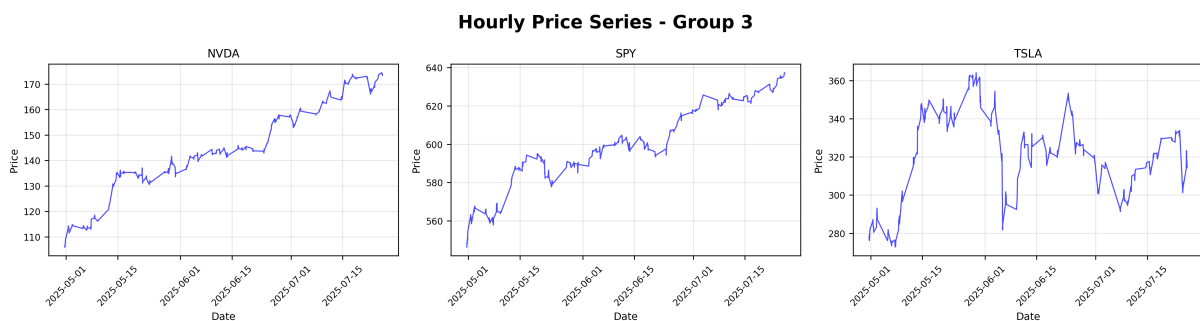


Figure 8: Evolution of Nvidia, Standard and Poor's 500 and Tesla price over time for hourly frequency

cause it ensures mathematical and statistical properties like time-additivity. The choice of logarithm of realized variance is backed by the rough volatility literature, it suggests that this process (log-RV) is best approximated by a Fractional Brownian Motion. [Gatheral et al. \(2018\)](#) and [Bennedsen et al. \(2022\)](#) argue that the log-volatility process displays near stationary behavior, strong autocorrelation and scaling laws consistent with fractional dynamics, in particular when $H < 0.5$. Logs here stabilize the variance of variance, and it avoids the boundedness and skewness of "raw" variance series, which are always positive and often heavy tailed. From a theoretical standpoint it aligns the data transformation with the structure of the model we wish to test: a rough process for $\log \sigma^2$. Log Realized Variance was computed by taking the log of 6.

5-Minute Price Series - Group 1

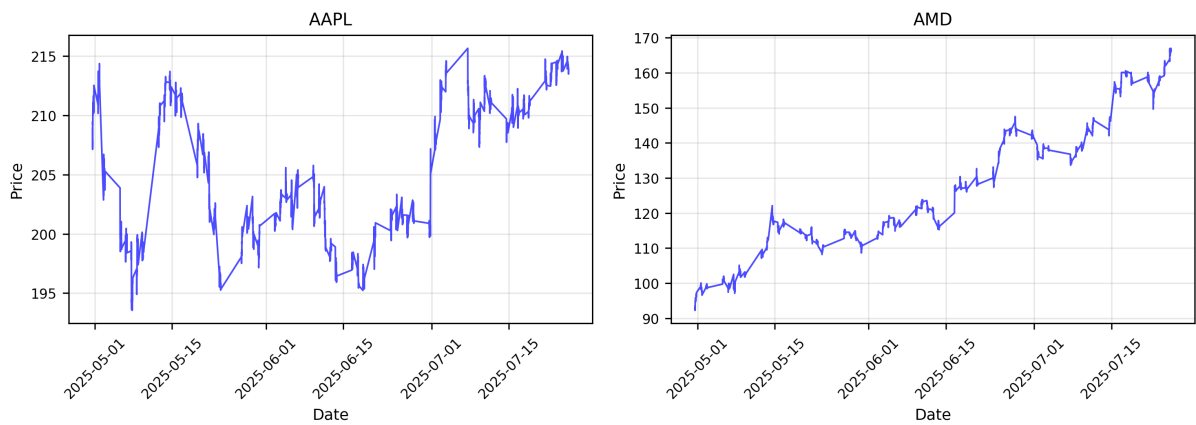


Figure 9: Evolution of Apple and AMD price over time for 5-minutes frequency

5-Minute Price Series - Group 2

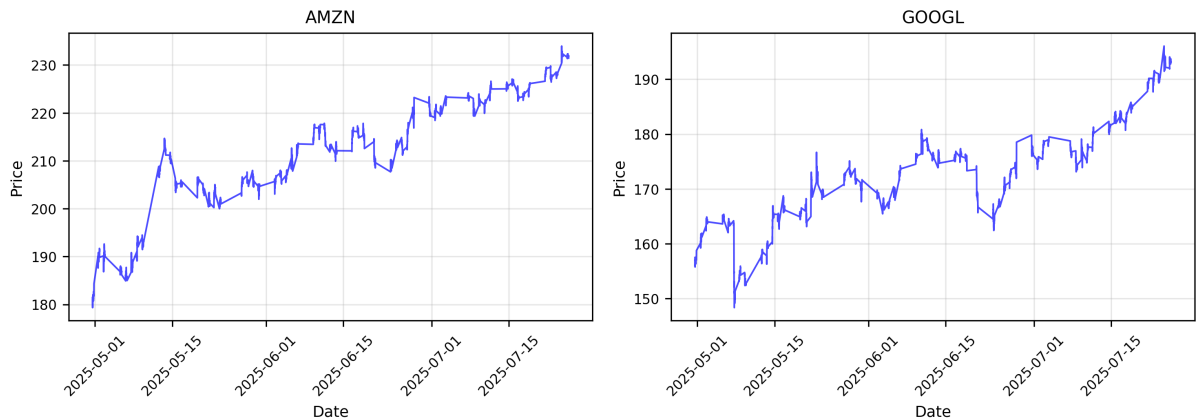


Figure 10: Evolution of Amazon and Google price over time for 5-minutes frequency

Even though we will address this potential issue in the next chapter, it is still relevant to the purpose of this one: choosing this time series intrinsically includes a potential timegrid flaw. The jumps that are inherent to the logic of trading days and trading calendar were not included in the simulation of the process. This issue is not addressed

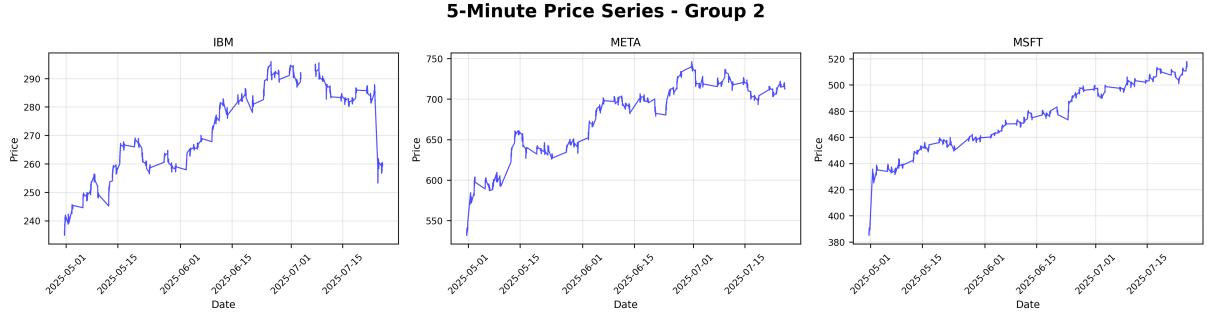


Figure 11: Evolution of IBM, Meta and Microsoft price over time for 5-minutes frequency

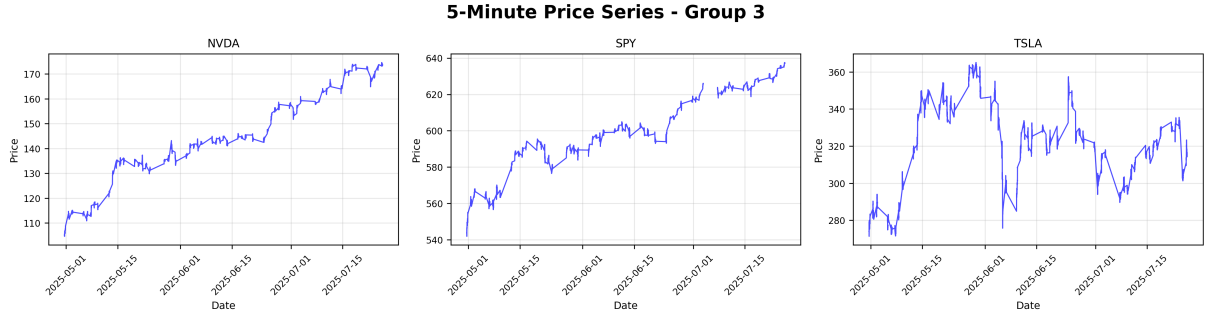


Figure 12: Evolution of Nvidia, Standard and Poor's 500 and Tesla price over time for 5-minutes frequency

directly because of two reasons: prior studies [Gatheral et al. \(2018\)](#); [El Euch and Rosenbaum \(2019\)](#) have shown that the roughness exponent H remains identifiable even at lower sampling frequencies, suggesting that the rough behavior is not purely a consequence of high-frequency market microstructure or weekend jumps, but a persistent characteristic of the log-volatility process itself, second, adopting other measures of how the volatility evolves at night-time or on weekends would require additional modeling assumptions that would introduce some more subjectivity for sure, whereas our current simplification preserves internal consistency, it is clear that the simulated paths are defined entirely by the fractional kernel and remain statistically interpretable. This topic will be discussed more in depth in the next chapter.

In order complement the diagnostics, we would like to test whether the series $\log-RV$ is stationary. We decided to implement the Augmented Dickey-Fuller (ADF) test for weak-stationarity, whether it fluctuates around a stable mean and has time-invariant second moments. This is crucial because it differentiates between non-stationary process that drift indefinitely and stationary ones which are stable, the latter are helpful to us because they further simplify the analysis. The ADF test is the main way to test for the presence of a unit root, thereby checking whether the series is nonstationary. The test is based on estimating the following regression:

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t, \quad (21)$$

where: y_t is the observed time series (in this case, $\log RV_t$), $\Delta y_t = y_t - y_{t-1}$ is the first difference, p is the number of lagged differences used to account for autocorrelation in the residuals, ε_t is a white noise error term. The key parameter is γ , which captures whether the lagged level of the series has explanatory power for its change. The test is framed as: null hypothesis $H_0 : \gamma = 0$ (non-stationary), whereas the alternative hypothesis is $H_1 : \gamma < 0$ (stationary). The test statistics and p-values computed are in the following table. This allows us to reject the null hypothesis at virtually any insightful significance levels, thereby the processes do not contain a stochastic trend and are weakly stationary.

Table 1: ADF statistics for each ticker and frequency.

Ticker	ADF (5min)	ADF (Daily)	ADF (Hourly)
AAPL	-16.594***	-4.536***	-6.398***
AMD	-16.576***	-5.061***	-6.720***
AMZN	-16.640***	-4.926***	-6.446***
GOOGL	-16.901***	-4.844***	-6.602***
IBM	-16.810***	-4.745***	-6.544***
META	-17.168***	-4.565***	-6.756***
MSFT	-17.003***	-4.478***	-6.154***
NVDA	-16.557***	-4.826***	-6.281***
SPY	-17.158***	-4.444***	-6.824***
TSLA	-16.587***	-4.837***	-6.464***

*** $p < 0.05$ (Accepted at 5% level)
** $p < 0.10$ (Accepted at 10% level)
* $p \geq 0.10$ (Not significant)

Therefore we can state that these same time series will later be used to calibrate and compare alternative volatility models under $\mathbb{P}$ with a somewhat solid empirical foundation.

4.2 Parameter Estimates

Before turning to model comparison in terms of forecasting accuracy, it is necessary to pause and examine the estimated parameters themselves, since these quantities encapsulate much of the structure that the models are trying to capture. The estimation strategy has been deliberately kept homogeneous across specifications, so that differences in results can be attributed more to the models than to the mechanics of estimation. For GARCH, parameters are obtained through quasi-maximum likelihood, which is the canonical approach given its recursive structure and has the advantage of delivering asymptotically consistent estimates under broad conditions. In the case of the HAR-RV model, estimation

is performed via rolling-window ordinary least squares, where the log realized volatility is regressed on its daily, weekly, and monthly lags. This allows coefficients to vary over time, thus reflecting changes in volatility persistence. Finally, for the RFSV specification, parameters such as the Hurst exponent H and the volatility-of-volatility coefficient ν are inferred through log-log regressions of structure functions, complemented by scaling-law diagnostics. This is consistent with the standard procedure in the rough volatility literature, where direct likelihood-based estimation is often intractable but moment-based methods have proven robust.

Throughout this research, forecasts and evaluation targets live on the log-realized-variance scale, $y_t = \log(\text{RV}_t)$. In particular, GARCH one-step-ahead variance forecasts—estimated via QML on $100\times$ log-returns—are converted to this display scale by dividing the model’s variance by 100^2 and taking logs, with a one-off continuity shift to anchor the first forecast to the last observed y_t . For HAR-RV, both the dependent variable and its predictors are y_t and its calendar time aggregates (daily, weekly, monthly), estimated via rolling OLS so that coefficients can vary over time. For RFSV, parameters are obtained from a weighted log-log regression of absolute increments to infer the Hurst exponent H and a volatility-of-volatility scale ν . forecasts are produced directly for y_t . For GARCH, α and β are absolute quantities and primarily inform persistence (via $\alpha + \beta$), while ω is scale-dependent because estimation is on $100\times$ returns, unconditional variance comparisons should therefore be mapped to the paper’s scale via $\log((\omega/(1 - \alpha - \beta))/100^2)$. In the case of HAR-RV, the coefficients $(c, \beta_d, \beta_w, \beta_m)$ are slopes on log-RV aggregates, hence dimensionless, but their absolute magnitude is not "universal" since it depends on the lag construction (calendar averages) and on the rolling window used at estimation. For RFSV, $H \in (0, 1)$ is scale-free and captures path roughness, whereas ν controls the amplitude of series’ fluctuations and is a scale parameter, so its absolute value is meaningful within a fixed sampling scheme but less comparable across frequencies or datasets. Absolute levels are not the object of interest. What is informative are relative patterns across assets, frequencies, or estimation windows that share the same preprocessing. Even without additional runs, one can rank assets by GARCH persistence ($\alpha + \beta$) and implied long-run log-variance, or compare HAR persistence allocation (daily vs. weekly vs. monthly weights) and ϕ . It is possible to compare RFSV roughness via H and the within frequency scale ν . If future runs change sample windows or frequency but keep the same log-RV target, these relative diagnostics remain directly comparable on the common y_t scale.

Looking at the results in general terms, it is possible to note that each family of models leaves its own signature on the parameter space. GARCH estimates typically feature a high degree of persistence, with α small but positive and β close to unity, so that the conditional variance evolves very smoothly and shocks are absorbed gradually. This is expected, since the model is designed to mimic volatility clustering via an autoregressive

mechanism in squared returns. HAR parameters instead distribute persistence across different horizons: the daily coefficient tends to be the largest, often in the 0.3–0.4 range, while weekly and monthly lags come in smaller but still statistically significant, capturing medium-term dependence. This allocation of weights mirrors the stylized fact that volatility shocks propagate over different horizons with diminishing strength. For RFSV, estimates of H are consistently in the rough regime, clustering around 0.1 to 0.15 at the daily frequency, which is strikingly close to the values reported in benchmark studies such as [Gatheral et al. \(2018\)](#). At higher frequencies, estimates move upward toward 0.2 or slightly more, a pattern that is well documented and generally attributed to microstructure noise inflating the perceived smoothness of the process. The second parameter of interest, ν , which controls the scale of volatility fluctuations, shows a stable pattern across assets, suggesting that the intensity of log-volatility innovations is a structural property rather than an artefact of the sampling frequency.

Turning now to each model in greater detail, starting with GARCH, the α parameters estimated on daily data indicate that the model assigns only a modest role to immediate shocks, with values typically around 0.05–0.1, while the β parameters range from 0.8 to 0.9, confirming that persistence is carried almost entirely by the autoregressive term. The ω parameter, which governs the unconditional variance, varies across stocks but remains relatively small, reflecting the scale of the underlying series. This triplet of parameters produces a variance dynamics that is smooth and mean-reverting, but as anticipated, when the same procedure is attempted at the 5-minute horizon the model breaks down, either failing to converge or producing unstable estimates. This underscores how GARCH is intrinsically a daily-frequency tool, ill-suited to capture the noisy and irregular structure of realized variance at high frequency.

Table 2: GARCH(1,1) Coefficients with Standard Errors

Ticker	Frequency	α	α SE	β	β SE	ω	ω SE
AAPL	daily	0.1061	0.0037	0.8190	0.0142	0.2925	0.0399
AAPL	hourly	0.0175	0.0032	0.7602	0.0484	0.0587	0.0118
AMD	daily	0.1274	0.0004	0.6455	0.0018	2.4378	0.0204
AMD	hourly	0.0000	0.0000	0.7391	0.0758	0.3605	0.1041
AMZN	daily	0.0930	0.0032	0.8061	0.0122	0.4997	0.0413
AMZN	hourly	0.0987	0.0079	0.1059	0.0588	0.1863	0.0122
GOOGL	daily	0.0172	0.0004	0.0000	0.0000	0.9999	0.0005
GOOGL	hourly	0.0977	0.0013	0.0000	0.0000	0.7122	0.0456
IBM	daily	0.0015	0.0002	0.0000	0.0000	0.9999	0.0017
IBM	hourly	0.0048	0.0003	0.0002	0.0010	0.9999	0.0002
META	daily	0.1001	0.0006	0.8691	0.0017	0.1901	0.0044

Continued on next page

Table 2 – continued from previous page

Ticker	Frequency	α	α SE	β	β SE	ω	ω SE
META	hourly	0.0866	0.0032	0.6989	0.0666	0.0867	0.0191
MSFT	daily	0.0115	0.0042	0.9685	0.0114	0.0603	0.0183
MSFT	hourly	0.0062	0.0024	0.9368	0.0136	0.0062	0.0013
NVDA	daily	0.1604	0.0044	0.6577	0.0011	2.7211	0.0665
NVDA	hourly	0.0303	0.0092	0.6425	0.0901	0.1899	0.0488
SPY	daily	0.1594	0.0004	0.8017	0.0017	0.0710	0.0017
SPY	hourly	0.0151	0.0025	0.9550	0.0094	0.0015	0.0005
TSLA	daily	0.0000	0.0000	0.9937	0.0011	0.1854	0.0235
TSLA	hourly	0.0000	0.0000	0.8028	0.0482	0.3922	0.0965

For the HAR-RV model, coefficient estimates confirm its interpretability as a distributed-lag approximation to long memory. The daily component dominates, consistent with the idea that volatility shocks are most persistent at the shortest horizon. Weekly lags add an intermediate layer of persistence, while monthly lags are weaker but non-negligible. Rolling estimation shows that these weights are not static: during periods of elevated market stress, the weekly coefficient tends to rise, indicating a lengthening of the memory structure. This flexibility is a strength of HAR, allowing it to accommodate regime changes without abandoning its linear form. Compared to GARCH, HAR is more capable of operating at multiple frequencies, as its regressions naturally adapt to the sampling scale.

Table 3: HAR-RV Coefficients with Standard Errors

Ticker	Frequency	Const	Const SE	Daily Lag	Daily SE	Weekly Lag	Weekly SE	Monthly Lag	Monthly SE
AAPL	5min	227.7021	9.4349	0.7770	0.0056	-0.0905	0.0356	10.8999	0.4602
AAPL	daily	-9.7047	0.0759	0.2567	0.0010	-1.8813	0.0029	1.9344	0.0065
AAPL	hourly	-48.1261	0.2086	0.7191	0.0011	-0.0021	0.0067	-2.0147	0.0100
AMD	5min	166.3137	7.1343	0.8387	0.0040	-0.1272	0.0192	8.1214	0.3424
AMD	daily	-16.2579	0.1405	0.2899	0.0007	-1.8161	0.0049	1.2886	0.0152
AMD	hourly	-47.1714	0.2665	0.7053	0.0010	0.0240	0.0062	-2.0124	0.0136
AMZN	5min	145.8784	5.8203	0.8506	0.0047	-0.1548	0.0209	7.0743	0.2811
AMZN	daily	-9.6727	0.0926	0.2649	0.0019	-1.9511	0.0015	1.9772	0.0094
AMZN	hourly	-54.9169	0.3093	0.7664	0.0013	-0.0540	0.0022	-2.3137	0.0147
GOOGL	5min	159.1783	9.5506	0.8502	0.0061	-0.1316	0.0192	7.7307	0.4669
GOOGL	daily	-13.9287	0.1309	0.2883	0.0012	-1.8557	0.0022	1.5396	0.0106
GOOGL	hourly	-48.8780	0.2909	0.7324	0.0009	-0.0427	0.0033	-2.0300	0.0146
IBM	5min	158.1015	9.1940	0.8442	0.0052	-0.1363	0.0247	7.6766	0.4467
IBM	daily	-11.4678	0.0553	0.3125	0.0007	-1.7424	0.0048	1.6139	0.0071

Continued on next page

Table 3 – continued from previous page

Ticker	Frequency	Const	<i>Const SE</i>	Daily Lag	<i>Daily SE</i>	Weekly Lag	<i>Weekly SE</i>	Monthly Lag	<i>Monthly SE</i>
IBM	hourly	-51.0645	<i>0.3117</i>	0.7116	<i>0.0010</i>	-0.0026	<i>0.0045</i>	-2.1403	<i>0.0164</i>
META	5min	276.4138	<i>8.9913</i>	0.7405	<i>0.0079</i>	-0.0964	<i>0.0390</i>	13.2422	<i>0.4371</i>
META	daily	-9.7870	<i>0.1469</i>	0.2689	<i>0.0013</i>	-2.0118	<i>0.0019</i>	2.0224	<i>0.0120</i>
META	hourly	-47.6714	<i>0.2476</i>	0.7500	<i>0.0007</i>	-0.0316	<i>0.0046</i>	-1.9988	<i>0.0128</i>
MSFT	5min	187.0282	<i>8.4924</i>	0.8283	<i>0.0050</i>	-0.0998	<i>0.0228</i>	8.8945	<i>0.4089</i>
MSFT	daily	-12.9261	<i>0.1054</i>	0.2379	<i>0.0009</i>	-2.0649	<i>0.0023</i>	1.9223	<i>0.0084</i>
MSFT	hourly	-48.6318	<i>0.2714</i>	0.7126	<i>0.0010</i>	-0.0002	<i>0.0053</i>	-2.0102	<i>0.0123</i>
NVDA	5min	301.0756	<i>9.5526</i>	0.6981	<i>0.0104</i>	-0.1220	<i>0.0322</i>	14.5118	<i>0.4711</i>
NVDA	daily	-14.3495	<i>0.0966</i>	0.2808	<i>0.0011</i>	-1.9040	<i>0.0039</i>	1.5232	<i>0.0068</i>
NVDA	hourly	-57.1578	<i>0.2278</i>	0.6559	<i>0.0008</i>	0.1348	<i>0.0060</i>	-2.5204	<i>0.0097</i>
SPY	5min	412.0355	<i>18.0146</i>	0.6023	<i>0.0107</i>	-0.0508	<i>0.0527</i>	19.3031	<i>0.8337</i>
SPY	daily	-12.9199	<i>0.0761</i>	0.2808	<i>0.0007</i>	-1.8589	<i>0.0033</i>	1.6997	<i>0.0053</i>
SPY	hourly	-57.2874	<i>0.1993</i>	0.7103	<i>0.0011</i>	0.0073	<i>0.0036</i>	-2.4015	<i>0.0086</i>
TSLA	5min	192.7594	<i>10.7677</i>	0.8152	<i>0.0103</i>	-0.1557	<i>0.0216</i>	9.4395	<i>0.5340</i>
TSLA	daily	-13.3106	<i>0.0666</i>	0.3201	<i>0.0007</i>	-1.7268	<i>0.0029</i>	1.3553	<i>0.0064</i>
TSLA	hourly	-47.5993	<i>0.3081</i>	0.7494	<i>0.0010</i>	-0.0542	<i>0.0033</i>	-1.9858	<i>0.0152</i>

The RFSV model provides perhaps the most revealing parameters. The estimates of H around 0.1 at daily frequency corroborate the rough volatility hypothesis, namely that volatility paths are far more irregular than a Brownian motion would suggest. At intraday scales, H increases toward 0.2–0.25, a finding consistent with the known upward bias caused by market microstructure effects. The ν coefficient, scaling the variance of the increments of log-volatility, displays a remarkable stability across assets and frequencies, which reinforces the interpretation of roughness as a universal structural feature rather than an idiosyncratic anomaly. These estimates place the RFSV model in a favorable position: it matches the empirical regularities documented in the literature and does so with a parsimonious parameter set. The fact that such values emerge so clearly from independent regressions across different stocks and frequencies strengthens the case for roughness as a fundamental property of volatility, echoing both the theoretical arguments of [Bayer et al. \(2016\)](#) and the more recent empirical contributions of [Bennedsen et al. \(2022\)](#).

Table 4: RFSV Parameters (H and ν) across Frequencies

Ticker	Frequency	H	ν
AAPL	5min	0.258254	1.301218
AAPL	daily	0.106999	8.978818
AAPL	hourly	0.140557	4.043971
AMD	5min	0.293177	1.451645
AMD	daily	0.134688	9.282837
AMD	hourly	0.138034	4.588386
AMZN	5min	0.259574	1.260872
AMZN	daily	0.120968	9.009120
AMZN	hourly	0.151203	3.713702
GOOGL	5min	0.314240	1.170323
GOOGL	daily	0.120791	9.031379
GOOGL	hourly	0.144357	4.017108
IBM	5min	0.293560	1.428934
IBM	daily	0.110283	8.800914
IBM	hourly	0.134366	3.996294
META	5min	0.226605	1.419021
META	daily	0.118314	9.027839
META	hourly	0.151834	3.931790
MSFT	5min	0.232887	1.391054
MSFT	daily	0.105501	8.612747
MSFT	hourly	0.143275	3.919674
NVDA	5min	0.287844	1.373913
NVDA	daily	0.129303	9.131088
NVDA	hourly	0.139918	3.988023
SPY	5min	0.245873	1.280417
SPY	daily	0.112345	8.923456
SPY	hourly	0.141224	3.981203

In conclusion, the parameter estimates across models highlight not only the technical adequacy of each specification but also the conceptual underpinnings of volatility modeling itself. GARCH reproduces clustering through near-unit persistence, HAR spreads memory across horizons in an interpretable way, and RFSV condenses irregularity and persistence into the roughness parameter H . The coherence of the RFSV estimates with established literature is especially notable, lending credence to its role as a benchmark candidate for modern volatility analysis.

5 Results and Conclusions

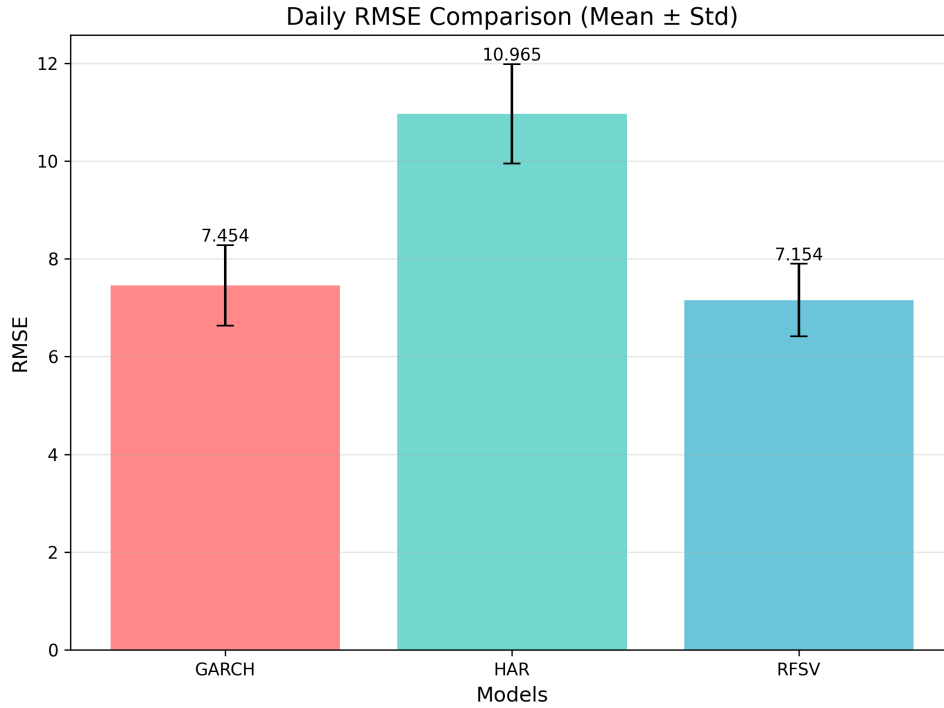
All forecasts and evaluations are carried out on the log–realized–variance scale, $y_t \equiv \log(\text{RV}_t)$, for a panel of large capitalization U.S. equities plus the S&P 500, observed at three sampling schemes: daily (July 2023–July 2025) and intraday at 1-hour and 5-minute over the most recent 60 trading days.¹ This choice ensures a common display scale across models and frequencies and follows the rough-volatility literature’s focus on log RV. For each frequency, the last $H = 25$ observations are held out for out-of-sample testing; the remaining observations form the in-sample set used for rolling re-estimation. At each out-of-sample date $\tau+h$ ($h = 1, \dots, H$) we re-fit the model on a trailing window of size $W = \min(\lfloor N_{\text{train}}/2 \rfloor, 250)$ (daily, hourly, 5-min), and issue a one-step-ahead forecast for $y_{\tau+h}$. For daily RFSV we fix $W = 100$ to stabilize the scaling regression used to infer H and ν . GARCH is not estimated at 5-minute due to instability and lack of statistical meaning at that horizon in this setting. The evaluation target $y_t = \log(\text{RV}_t)$ is common. GARCH one-step variance forecasts, estimated on $100 \times$ log-returns, are mapped to the y_t scale by dividing the conditional variance by 100^2 and taking logs, with a continuity shift to anchor the first forecast to the last observed y_t . HAR-RV is estimated directly on y_t and its calendar aggregates, and RFSV produces forecasts for y_t from the conditional Gaussian law of fBM. Rolling estimation reflects time variation in persistence without overfitting the out-of-sample segment. A fixed holdout period ($H = 25$) at each frequency yields the same forecast "difficulty" across models.

The deliberate focus on a homogeneous cluster of firms allowed the analysis to operate within a relatively coherent volatility regime, though at the cost of introducing sectorial bias, with stronger cross-correlation in return and volatility dynamics than would be expected in a broader market sample. Such dependence is not without drawbacks, yet it strengthens the interpretability of results by ensuring that model differences are not driven by structural heterogeneity across industries. Future work could fruitfully explore whether the conclusions reached here hold across different sectors or for a wider market index.

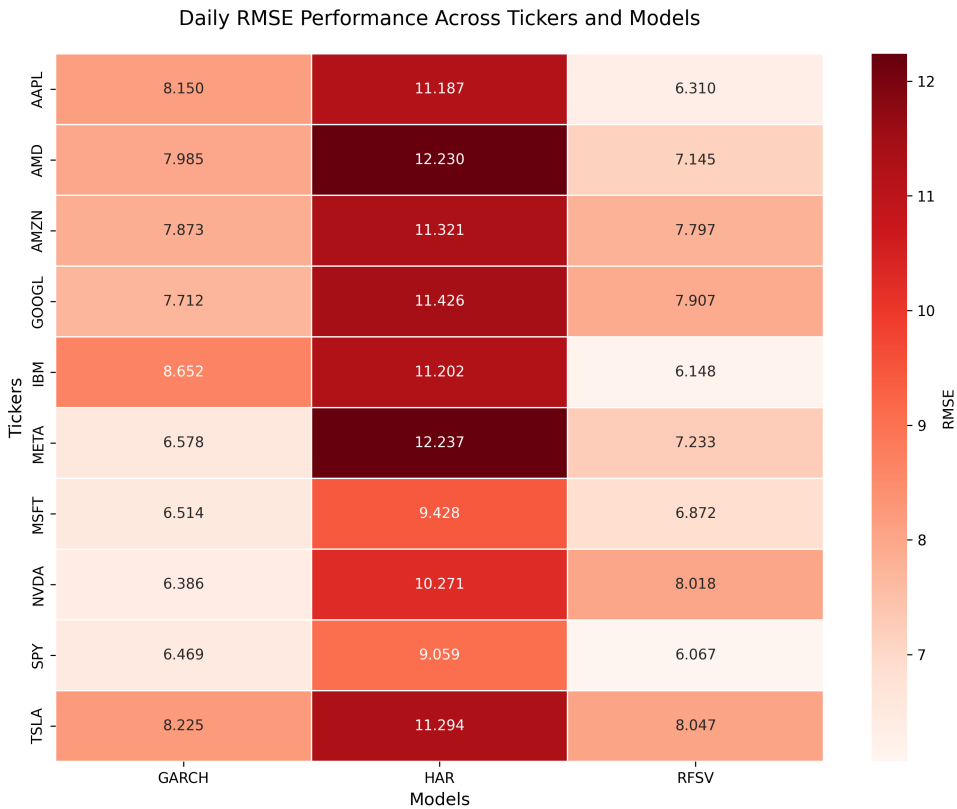
The comparative performance of the models, expressed in terms of RMSE, is illustrated in Figures 13b–15a. These plots summarize both per-ticker results (heatmaps) and aggregated averages with dispersion (barplots), providing a dual perspective on robustness and variability across firms.

For GARCH, the evidence is broadly consistent with expectations. At the daily frequency the model is still competitive, though never leading, as shown in Figure 13b. At higher frequencies its limitations become evident: persistence close to unity and reliance on squared return dynamics fail to accommodate the irregular structure of realized volatility. This is reflected in the hourly comparison (Figure 14b), where GARCH systematically

¹See Data subsection for the exact sample spans and motivation.



(a) Forecast model comparison, daily frequency.



(b) RMSE heatmap, daily frequency.

Figure 13: Comparison of model performance at the daily frequency. Panel (a) reports forecast accuracy across models, while panel (b) shows the RMSE distribution across tickers in heatmap form.

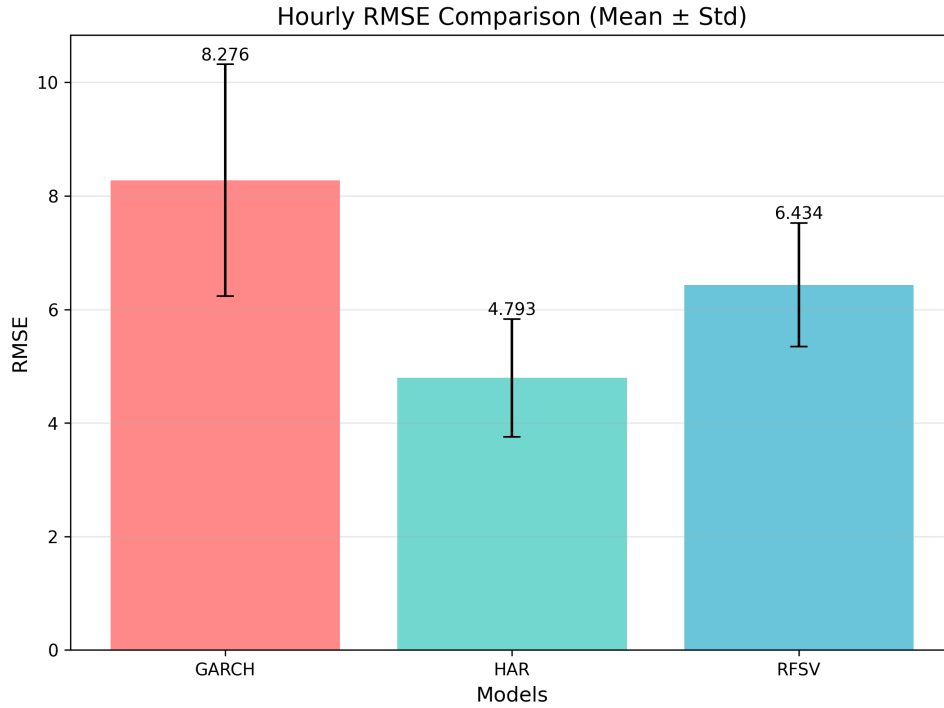
records the highest errors, and even more so in the 5-minute setting, where the model could not be estimated in a statistically meaningful way. The interpretation is straightforward: GARCH remains serviceable at low frequency, but its framework does not scale when confronted with more granular data.

The HAR-RV model demonstrates a more balanced profile. At the hourly scale (Figure 14b) it performs particularly well, benefitting from its distributed-lag structure that allows persistence to be spread across horizons. While it underperforms at the daily level relative to both GARCH and RFSV, and is not dominant at the 5-minute frequency, HAR nonetheless shows a capacity to adapt across contexts without breaking down. This robustness is visible in the barplots (Figures 13a and 14a), where its average error remains consistently moderate. It is therefore fair to describe HAR as a reliable intermediate option, capable of capturing medium-term memory even if it does not fully exploit the roughness present at high frequency.

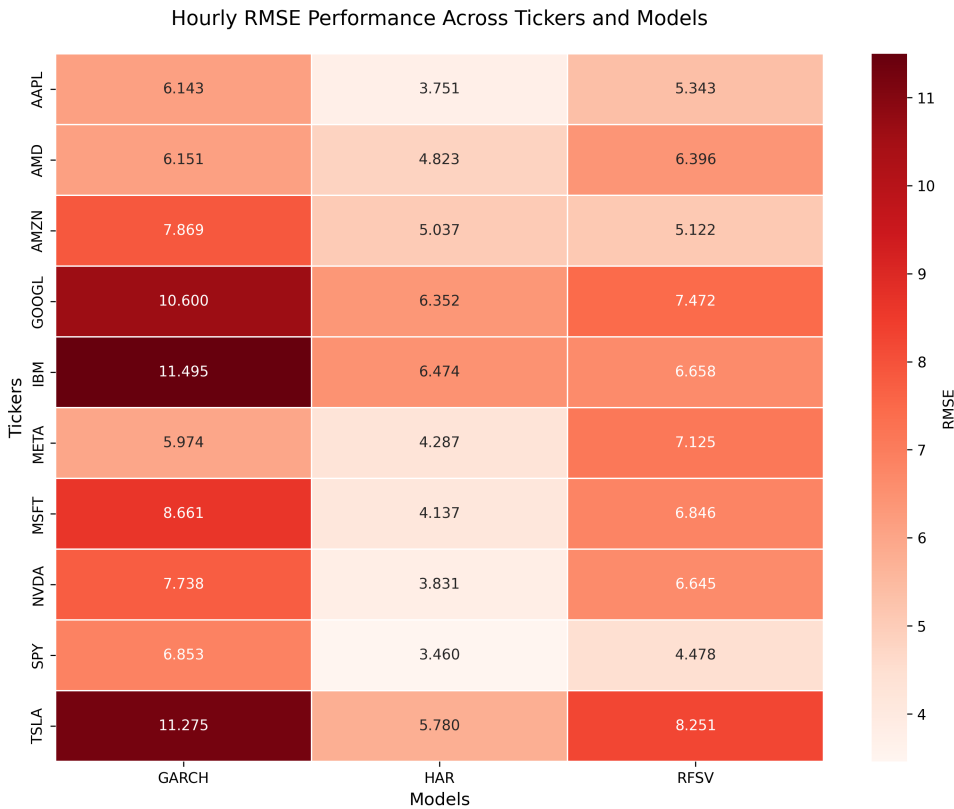
The RFSV model, finally, emerges as the most promising among the three. At the daily frequency it matches or surpasses GARCH (Figure 13b), delivering lower RMSE for nearly all tickers. At the hourly scale it competes closely with HAR, and at the 5-minute frequency (Figure 15b) it establishes a clear advantage, confirming the theoretical expectation that rough volatility models are particularly well suited for high-frequency environments. Beyond average errors, what distinguishes RFSV is the consistency of its performance across firms and across scales: it never collapses in the way GARCH does, and it never becomes structurally inferior, as HAR sometimes does at longer horizons.

Taken together, the empirical analysis suggests that forecasting volatility remains an intrinsically context-dependent exercise. Traditional GARCH models are not obsolete, but their scope is increasingly narrow. HAR continues to offer a tractable and interpretable solution, especially for intermediate horizons. RFSV, however, captures both the fine structure and the persistence of volatility in a way that is theoretically grounded and empirically validated. Its parsimony, transparency, and adaptability make it a serious candidate to succeed GARCH as a benchmark model in financial econometrics.

This conclusion is not to be read as a definitive settlement of the matter. As volatility modeling continues to evolve, one can imagine extensions that combine roughness with other structural elements, such as multifactor kernels or time-varying parameters, in the spirit of the hybrid schemes introduced in the literature. Yet the evidence collected here provides a strong basis for arguing that roughness is not an artifact of data or estimation, but a genuine structural property of financial volatility. In that sense, RFSV does not merely outperform on a metric: it offers a lens through which volatility itself may be more faithfully understood.

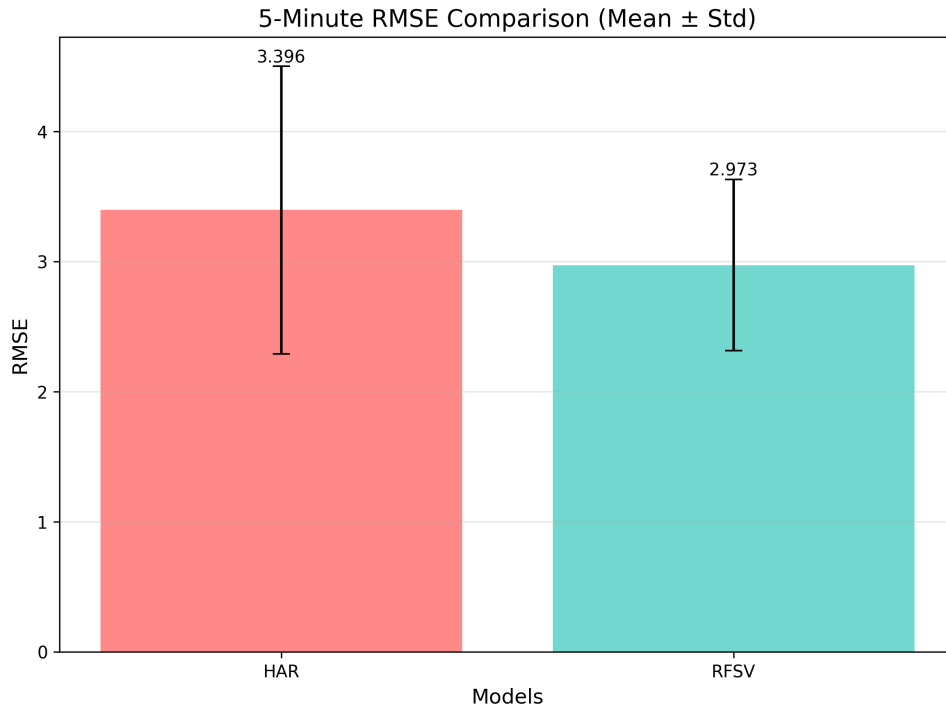


(a) Forecast model comparison, hourly frequency.

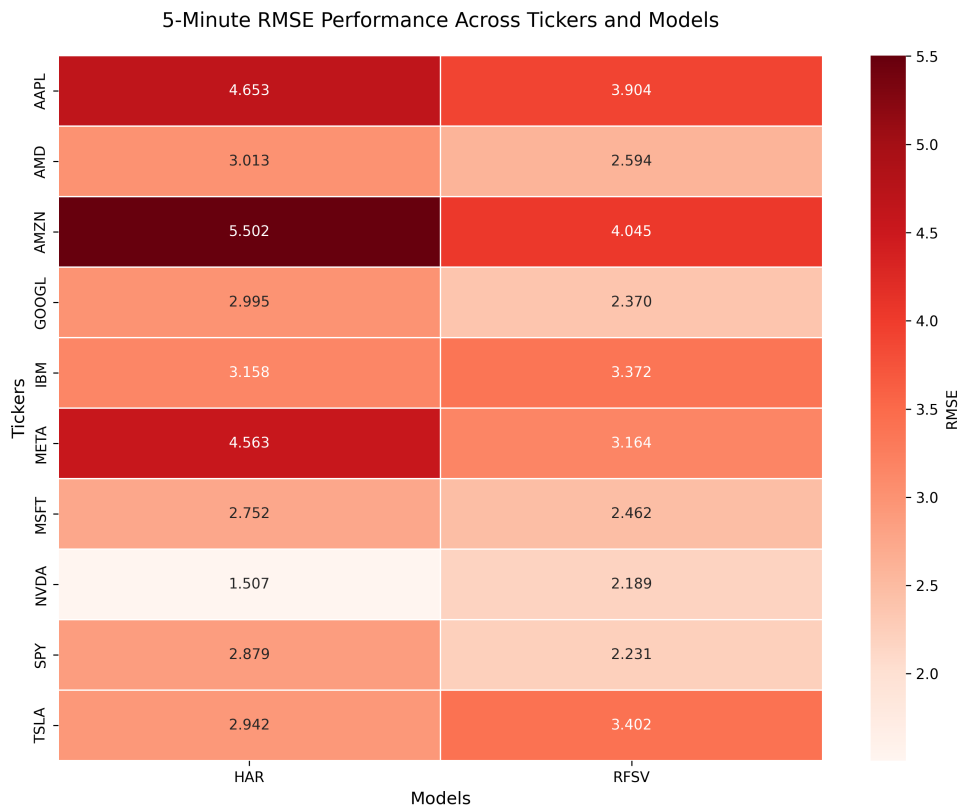


(b) RMSE heatmap, hourly frequency.

Figure 14: Comparison of model performance at the hourly frequency. Panel (a) reports forecast accuracy across models, while panel (b) shows the RMSE distribution across tickers in heatmap form.



(a) Forecast model comparison, 5-minute frequency.



(b) RMSE heatmap, 5-minute frequency.

Figure 15: Comparison of model performance at the 5-minute frequency. Panel (a) reports forecast accuracy across models, while panel (b) shows RMSE distribution across tickers in heatmap form.

References

- Abi Jaber, E. and El Euch, O. (2019). Multifactor approximation of rough volatility models. *SIAM Journal on Financial Mathematics*, 10(2):309–349.
- Baillie, R. T., Bollerslev, T., and Mikkelsen, H. O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of econometrics*, 74(1):3–30.
- Bayer, C., Friz, P., and Gatheral, J. (2016). Pricing under rough volatility. *Quantitative Finance*, 16(6):887–904.
- Bennedsen, M., Lunde, A., and Pakkanen, M. S. (2017). Hybrid scheme for Brownian semistationary processes. *Finance and Stochastics*, 21(4):931–965.
- Bennedsen, M., Lunde, A., and Pakkanen, M. S. (2022). Decoupling the short-and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5):961–1006.
- Bibinger, M., Yu, J., and Zhang, C. (2025). Modeling and forecasting realized volatility with multivariate fractional Brownian motion. *arXiv preprint arXiv:2504.15985*.
- Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *The Review of Economics and Statistics*, pages 542–547.
- Comte, F. and Renault, E. (1998). Long memory in continuous-time stochastic volatility models. *Mathematical Finance*, 8(4):291–323.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.
- Davies, R. B. and Harte, D. S. (1987). Tests for Hurst effect. *Biometrika*, 74(1):95–101.
- Dragulescu, A. A. and Yakovenko, V. M. (2002). Probability distribution of returns in the heston model with stochastic volatility. *Quantitative Finance*, 2(6):443.
- El Euch, O. and Rosenbaum, M. (2018). The microstructural foundations of leverage effect and rough volatility. *Finance and Stochastics*, 22(2):241–280.
- El Euch, O. and Rosenbaum, M. (2019). The characteristic function of rough Heston models. *Mathematical Finance*, 29(1):3–38.
- Engle, R. F. and Bollerslev, T. (1986). Modelling the persistence of conditional variances. *Econometric reviews*, 5(1):1–50.
- Fukasawa, M. (2017). Short-time at-the-money skew and rough fractional volatility. *Quantitative Finance*, 17(2):189–198.

- Fukasawa, M. (2021). Volatility has to be rough. *Quantitative finance*, 21(1):1–8.
- Gatheral, J., Jaisson, T., and Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6):933–949.
- Hansen, P. R. and Lunde, A. (2005). A forecast comparison of volatility models: does anything beat a GARCH (1, 1)? *Journal of Applied Econometrics*, 20(7):873–889.
- Mandelbrot, B. B. and Van Ness, J. W. (1968). Fractional Brownian motions, fractional noises and applications. *SIAM review*, 10(4):422–437.
- McFadden, D. (1989). A method of simulated moments for estimation of discrete response models without numerical integration. *Econometrica: Journal of the Econometric Society*, pages 995–1026.
- Reschenhofer, E. (2013). Does anyone need a GARCH (1, 1). *Journal of Finance and Accounting*, 1(2):48–53.
- Shevchenko, G. (2014). Fractional Brownian motion in a nutshell. *arXiv preprint arXiv:1406.1956*.